

# Shared SDG Labels Don't Mean Shared Framing — Embedding-Based Register-Topic Decomposition of Research-Policy Divergence Across the SDGs

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## Abstract

Shared SDG labels do not establish that research and policy *frame* a goal similarly (Armitage et al., 2020). This paper contributes a measurement framework that separates research-policy divergence into two independent dimensions: coverage (which SDGs each corpus prioritises) and within-SDG semantic framing (how differently the same SDG is discussed); framing is itself a composite signal of topic (substantive content) and register (communicative conventions), and this paper isolates topic from register using Iterative Nullspace Projection (INLP). Using frozen Sentence-BERT embeddings and a supervised logistic-regression classifier trained on 52,835 SDG-labelled reference segments (test macro- $F_1 = 0.816$ ), this study scores 3,105,144 segments from 2,536,771 AI-for-sustainability abstracts and 40,597 segments from 6,367 institutional policy sources, and measures both dimensions across all 17 SDGs.

The supervised classifier reproduces the coverage ranking established by prior keyword- and citation-based inventories: SDG 3 dominates research while SDGs 13, 16, and 17 dominate policy.

The raw coverage–semantic-gap correlation is nevertheless null ( $\rho = -0.012$ , permutation  $p = 0.959$ ). Decomposing within-SDG distance into topic and register components, however, reveals a systematic cancellation: the adjusted semantic gap is positively associated with coverage ( $\rho = 0.544$ ,  $p = 0.025$ ) while the removed register component is negatively associated ( $\rho = -0.390$ ,  $p = 0.120$ ). All  $p$ -values are two-sided Monte Carlo permutation  $p$ -values (100,000 resamples). The linkage operates asymmetrically: policy coverage is the strongest directional predictor (positive in 9/9 encoder–classifier configurations;  $\rho = +0.589$  for MPNet LR), while research coverage shows no consistent signal. Across the full  $n = 17$  SDG framework, only the adjusted association reaches significance; the register component is directionally negative but not significant, so the null raw gap reflects a partial, not complete, cancellation.

The result is nonetheless stable: the adjusted semantic-gap ranking survives every sensitivity check (encoders, classifier family, retrieval strategy, segment cap, and policy-source family) and distribution-aware metrics confirm it is a mean-direction effect. Coverage and framing should be measured and reported as separate axes of divergence; shared labels are insufficient evidence of shared framing (Heilinger et al., 2024).

**Keywords:** research-policy alignment; Sustainable Development Goals; semantic similarity; transformer embeddings; measurement protocol; bibliometrics.

**Data and code availability.** The analysis code and the frozen, processed outputs are version-controlled at <https://github.com/qmanhbeo/dissertation-bham>. The repository is private during thesis examination and will be made public after grade release; raw upstream data are retrieved from the OpenAlex API under its terms of use.

**Competing interests.** The author declares none.

# 1 Introduction

Research-policy alignment is often inferred from whether scientific and policy texts address the same topics (Bornmann et al., 2016), yet shared coverage does not establish comparable framing: a paper and a policy document can both invoke SDG 13 (Climate Action) or SDG 3 (Good Health) while emphasising different problems, actors, and interventions. High-level thematic agreement can therefore mask deeper divergence in how each discourse community approaches the same goal. Legislative agendas, ESG disclosures, and funding decisions increasingly treat SDG tags as evidence of alignment (Heilinger et al., 2024; Heras-Saizarbitoria et al., 2022); this assumption is empirically unfounded.

Existing methods for mapping research to the Sustainable Development Goals (SDGs) operate mainly at the level of lexical or bibliometric correspondence: keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). They measure whether a text is *about* a given SDG, not how it frames it. Scientometrics offers scale but lacks linguistic nuance; policy text analysis offers nuance but lacks computational scale (**Bornmann2016policy**; Toney-Wails et al., 2024). What has been missing is a method that bridges both, operating at scale while distinguishing register from topic. That interchangeability is what this paper questions. The coverage distribution reproduces the ranking those methods established, SDG 3 dominant in research and SDGs 13, 16, and 17 in policy, anchoring external validity before the paper turns to framing.

This paper proposes that research-policy alignment has at least two separable dimensions: coverage divergence (which SDGs each corpus prioritises) and semantic framing divergence (how differently they discuss the same SDG). Its contribution is a measurement framework. This framework advances AI-SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed textual-semantic proximity in embedding space; Section 2.2), treating semantic proximity as a register-sensitive proxy for framing, not evidence of substantive alignment. The framework operationalises both dimensions across all 17 SDGs with three frozen sentence-transformer encoders (MPNet, MiniLM, SciBERT; Section 3.3) and a supervised logistic-regression classifier, stress-tested against encoder, assignment method, policy source, and segment-cap choices, with a novel SDG-stratified INLP adaptation that separates register from topic by training the corpus classifier on balanced topic-corpus strata at each iteration.

The near-zero raw coverage-vs-semantic-gap correlation may reflect cancellation: topic divergence rises with coverage divergence while register divergence falls with it, and the two appear to offset (permutation  $p = 0.025$  and  $p = 0.120$  respectively,  $n = 17$

SDGs); the directional decomposition further suggests this linkage operates primarily through policy coverage, not research coverage. This matters because SDG tagging is now institutionalised: OpenAlex labels works, funders count SDG outputs, and journals print SDG badges (Heilinger et al., 2024; Heras-Saizarbitoria et al., 2022); if shared labels mask a register–topic cancellation, that threatens how the SDG-monitoring ecosystem reads its own signals. After register removal, the adjusted semantic gap identifies SDG 17 as the most divergent goal and the one most sensitive to how divergence is measured; under the raw gap (naive baseline), SDG 3 leads instead. SDG 15 is the stably least-divergent goal under both.

The central research question is: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?*

## 2 Literature Review

### 2.1 AI and the SDGs: What Bibliometrics Shows

Existing assessments of AI and the SDGs rest on expert views or publication counts rather than on the texts themselves (Vinuesa et al., 2020). Broader SDG scientometrics and the AI-specific inventories converge on one robust finding: SDG 3 (Good Health) and SDG 7 (Affordable and Clean Energy) are the most addressed goals in the AI-specific studies, followed by SDG 4 (Quality Education), SDG 13 (Climate Action), SDG 11 (Sustainable Cities and Communities), and SDG 16 (Peace, Justice and Strong Institutions). The evidence base spans a citation-based corpus of roughly 25,000 publications on the Millennium and Sustainable Development Goals (M&SDGs) (Bautista-Puig et al., 2021), 20,511 AI publications (2001–2020) (Singh et al., 2024), 108 AI-for-social-good projects (Cowls et al., 2021), and 1,288 big-data and AI publications (Nedungadi et al., 2024), together with a recent review of 792 AI applications in SDG-related research (Gohr et al., 2025). None of these studies measures within-SDG semantic framing.

SDG publication mapping is methodologically contingent. Armitage et al. (2020) shows that what counts as SDG-relevant depends on interpretive choices about queries, labels, and databases; Kashnitsky et al. (2024) reach the same caution from a broader comparison of Boolean, machine-learning, and hybrid mapping systems, finding no single best approach and no single “golden” validation set. The dependence persists in AI-based mapping: keyword baselines still omit publications that never state an SDG keyword explicitly, so the reported corpus boundary remains query-dependent (Yin et al., 2025). The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but its results remain conditional on corpus construction, centroid provenance, and model

choice.

The SDGs themselves are interdependent by design, yet no operational framework specifies how (Nilsson et al., 2016); single-SDG assignment is therefore a simplifying assumption. One goal breaks the pattern. Le Blanc (2015), in a network analysis of the proposed SDGs, treats SDG 17 (Partnerships for the Goals) as the dedicated goal for cross-cutting means of implementation: finance, trade, technology transfer, and capacity building, excluding it from cross-goal mapping because its targets are implementation infrastructure rather than substantive domains. The Griggs et al. (2017) guide likewise describes SDG 17 as the main enabler of the whole framework. Unlike every other SDG, whose discourse centres on a shared substantive topic, SDG 17 policy discourse concerns coordinating and resourcing implementation itself, a conceptual space that technical AI research abstracts are not built to occupy.

## 2.2 A Three-Level Framework of Research-Policy Alignment

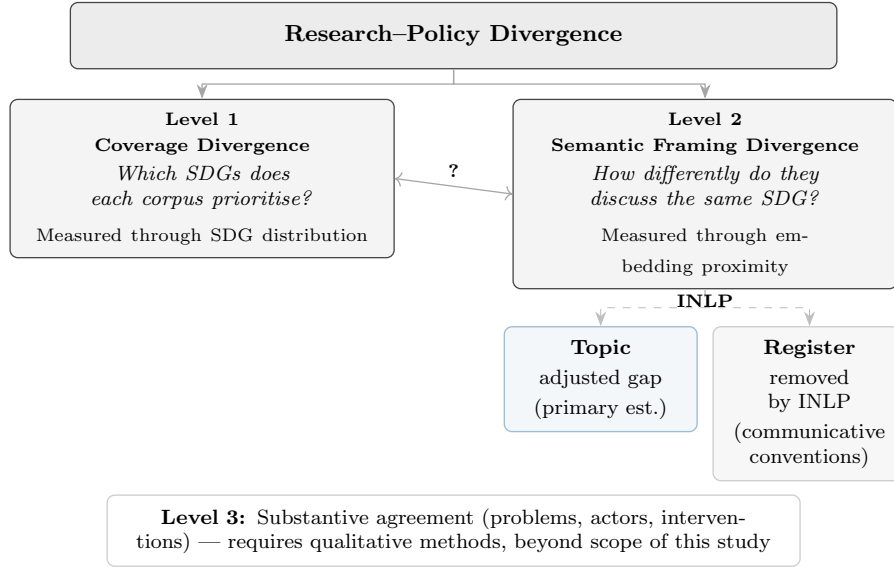
This paper proposes that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

**Level 1: lexical or bibliometric correspondence.** The dominant mode of existing SDG-mapping work: assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). Necessary but weak: shared vocabulary does not guarantee shared meaning (Bornmann et al., 2016), and keyword methods are brittle to vocabulary and database choices (Armitage et al., 2020).

**Level 2: observed textual-semantic proximity in embedding space.** Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers & Gurevych, 2019), and encoder-based SDG association has been benchmarked as a strong baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss. The present study is positioned at this level, and the limits of that measurement are developed in Section 2.4 below.

**Level 3: substantive comparison of implied problems, actors, and interventions.** Whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study’s concrete methodological contribution; Level 3 names the boundary it deliberately does not cross. Figure 1 illustrates this decomposition.



**Figure 1.** Conceptual framework: research-policy divergence separated into coverage and semantic framing dimensions, positioned within the three-level alignment construct.

## 2.3 Research-Policy Alignment in AI and Sustainability

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires: corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the deployment impacts governance must address (Strauss et al., 2025); global AI metrics emphasise technical and economic performance while policy discourse stresses ethical and social dimensions aligned with SDG concerns (Sioumalas-Christodoulou & Tympas, 2025); and research and policy communities can share trustworthy-AI terminology while differing in term salience and focal concerns (Toney-Wails et al., 2024). The pattern extends beyond AI: in conservation science, 59% of policy-prioritised research topics appear in fewer than 15 papers (McCarthy et al., 2026).

Adjacent scientometric work measures research-policy connection through policy-document mentions rather than semantic framing: only 1.2% of 191,276 climate-change publications have at least one policy mention (Bornmann et al., 2016).

Theory explains why such divergence is the default rather than a failure. Knowledge systems support action when they produce information that is simultaneously salient, credible, and legitimate (Cash et al., 2003). The epistemic-communities framework explains why shared framing matters for policy coordination under uncertainty: expert communities influence policy by articulating cause-effect relationships and framing issues for collective debate (Haas, 1992). Haas’s concept is broader than semantic proximity, involving shared normative and causal beliefs, validity criteria, and a common policy enterprise; the present study therefore does not operationalise an epistemic community directly, but measures



only the weaker condition of observed semantic proximity. Even this weaker condition is only a possible precondition for knowledge transfer: semantic proximity cannot show the existence, strength, or influence of an expert community (Cross, 2013). The two-communities theory holds that researchers and policy makers live in separate worlds with different values, reward systems, and languages, so divergence is the default state of research-policy interaction (Caplan, 1979).

The benchmark itself is not neutral. The SDG framework is a politically negotiated instrument with documented limitations: goals can create incentives to over-focus on target achievement (Fukuda-Parr, 2016), summit documents remain state-centric and avoid causal responsibility (Bexell & Jönsson, 2017), and Goal 8’s 3% annual growth target is empirically incompatible with the resource-use and emissions goals (Hickel, 2019). Reported SDG engagement is frequently superficial, so reported alignment should not be conflated with substantive commitment (Heilinger et al., 2024; Heras-Saizarbitoria et al., 2022). To the extent that research follows Mode 1 knowledge production (problems set within academic disciplines) rather than Mode 2 (problems arising in contexts of application) (Gibbons et al., 1994), divergence is not uniformly problematic: where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment.

## 2.4 Semantic Methods: Rationale and Cautions

The distributional hypothesis, that meaning can be described through patterns of linguistic co-occurrence (Harris, 1954), underlies the use of Sentence-BERT (Reimers & Gurevych, 2019) and related transformer encoders for large-scale semantic comparison. Applying them to compare two distinct discourse corpora, research abstracts and policy documents, is a novel usage rather than a directly benchmarked one, and the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform among the strongest as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length. Rodriguez et al. (2023)’s conText framework provides a computational-social-science precedent for comparing meaning across contexts, adopting a narrow, structuralist view of meaning as distributional co-occurrence and making no causal claims about human understanding. This study follows that same descriptive premise: semantic proximity is a distributional measurement, not evidence of substantive framing alignment. The implication is that embedding distance mixes topic, register, and contextual convention into a single number: distributional representations are not topic-only, since language models inherently encode varieties of language including register (Grieve et al., 2025), and stylistic variation is demonstrably present in distributional vector

spaces (Niu & Carpuat, 2017). This motivates the decomposition into these components (Section 3.8). McCarthy et al. (2026) supply a closer empirical precedent, showing with a multi-target SciBERT classifier that research–policy coverage gaps are measurable with transformer encoders outside the AI–SDG setting; this study extends that logic from a single conservation domain to the full SDG landscape, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition adds a further caution. Biber’s multidimensional (MD) analysis identifies six co-varying dimensions of linguistic variation across 23 spoken and written genres, derived from factor analysis of 67 linguistic features (Biber, 1988). Each dimension captures a continuum of co-occurring linguistic features rather than a binary speech/writing split (Biber, 1988; Biber & Conrad, 2009). Register is therefore a multi-layered property: two texts can differ on some dimensions and converge on others. The INLP procedure in this study removes a *linear* subspace from the embedding, which is a projection onto one or more directions in the 768-dimensional space; this is best understood as capturing the dominant linear component of Biber’s continuous register space, not as a claim that register is binary or that a single projection exhausts it. For the present study this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function; embedding distance is therefore treated as observed textual-semantic proximity, not a clean measure of topic alone.

## 2.5 Research Gap

To the best of the evidence reviewed, the literature reviewed above has not produced: (i) a systematic comparison of how academic AI-for-sustainability research and institutional SDG policy discourse differ in SDG coverage and within-SDG semantic framing; prior work measures research coverage, citation patterns, or policy-document uptake, but does not compare the two corpora directly on these dimensions; or (ii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. This framework is deliberately diagnostic: it maps where observed divergence is visible, without classifying whether any given gap reflects productive or problematic disconnection; the latter would require qualitative follow-up beyond the scope of embedding analysis.

These two contributions are motivated by the theoretical frameworks reviewed in Section 2.3. The two-communities theory holds that researchers and policymakers occupy separate worlds with different values, reward systems, and languages (Caplan, 1979). If these communities also develop different topical priorities, as the empirical record of research-policy divergence suggests (McCarthy et al., 2026; Strauss et al., 2025), then coverage divergence and semantic divergence should co-occur: SDGs where one community dominates attention should also show divergent framing, because the community

shaping the discussion brings its own communicative norms. This motivates H1a. The epistemic-communities framework further distinguishes the *direction* of dominance: where research prioritises a given SDG, the framing should reflect academic norms of knowledge production; where policy prioritises it, the framing should reflect institutional norms of coordination (Haas, 1992). H1b–H1d decompose the coverage-semantic relationship into these directional components. The Mode 1/Mode 2 distinction (Gibbons et al., 1994) provides a theoretical reason why not all divergence is failure: where research follows discipline-driven Mode 1 priorities while policy discourse follows contexts of application, coverage-semantic distance may reflect productive exploration rather than misalignment.

## 2.6 Research Questions and Hypotheses

The research question is operationalised by four sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs:

- **H1a** (motivating hypothesis): coverage gap  $\leftrightarrow$  semantic gap;
- **H1b**: research–policy dominance  $\leftrightarrow$  semantic gap;
- **H1c**: research coverage  $\leftrightarrow$  semantic gap;
- **H1d**: policy coverage  $\leftrightarrow$  semantic gap.

SDG 4 is treated as artefact-affected and examined separately (Appendix C.1). The operationalization of these tests, Pearson  $r$  and Spearman  $\rho$  between each predictor and semantic-gap scores including the  $n = 17$  scope conditions, is described in Section 3.9.

## 3 Methodology

### 3.1 Research Corpus

This study uses publicly available text data and requires no ethical approval.

The research corpus consists of 2,536,771 research abstracts retrieved from the OpenAlex API (Priem et al., 2022), covering publications between 2018 and 2025. Retrieval combined OpenAlex’s native SDG work filter with four high-precision AI search terms (**artificial intelligence, machine learning, deep learning, neural network**), giving 68 queries (17 SDGs  $\times$  4 terms). The term set draws on established practice for delimiting AI corpora (Hellwig et al., 2019; OECD, 2024), capturing a focused subset rather than the full universe of AI-related terms. Because this study measures semantic alignment between research and policy rather than counting publications, retrieval favours precision: each search term must clearly identify AI-related work, accepting that some relevant papers may be missed, to keep non-AI papers out of the embedding space. Complete query parameters are documented in the Retrieval and Query Chain (Appendix A).

A direct test re-ran retrieval using field-of-study membership for the same two AI fields (OECD, 2024) and re-scored the resulting 100,000-paper corpus through the identical pipeline. Coverage and semantic-gap rankings are stable under this alternative strategy (Section 4.4; Appendix B).

An important asymmetry follows: the research corpus is a tight intersection of AI and SDG content ( $AI \cap SDG$ ), while the policy corpus (Section 3.2) is a broader union covering institutional SDG discourse and AI governance. Following Armitage et al. (2020), the retrieval universe is a method-dependent corpus boundary. After deduplication and removal of records lacking usable abstracts, 2,536,771 abstracts were retained and tokenised into 3,105,144 segments (1.224 segments per abstract on average; 17.18% exceed the 374-token encoder window and split into multiple segments). A small number of records fail the 20-word segment minimum and are excluded: 2,543,698 records survive preprocessing, of which 6,927 are dropped at segmentation because their combined text is too short (mostly very short abstracts). This is a documented scope boundary, and the dropped share is below 0.3%. The corpus is predominantly English but no hard language filter is applied: roughly 0.5% of abstracts are predominantly non-English and are retained. The semantic-gap ranking reproduces under subsampling to 10,000 papers (Appendix F), two orders of magnitude larger than the non-English share. No balancing by SDG or citation count is applied.

### 3.2 Policy Corpus

The policy corpus contains 40,597 text segments from three collections, representing 6,367 unique source documents. It should be read as mixed *international institutional SDG policy discourse* rather than the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (13,123 segments, 94 documents): UN, OECD, EU, UK, and AU AI governance frameworks; IPCC assessment reports; SDSN sustainable development reports; and national AI strategies.
2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (21,346 segments, 4,225 documents): National and local self-assessments submitted to the UN High-Level Political Forum.
3. **UN General Debate Corpus** (Jankin et al., 2024) (6,128 segments, 2,048 documents): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024); paragraph-level keyword filtering precedes segmentation, so only SDG-relevant passages enter the pipeline, which is why its 6,128 segments are far fewer than the full corpus of speeches.

After cross-source deduplication by exact text match (SDGi retained first), 40,597 unique segments remained. The curated sub-corpus (94 documents) is the more relevant reference

for AI-governance-specific interpretation; Appendix I.6 reports a source-family sensitivity check.

### 3.3 Embedding Model and Normalisation

All texts were encoded with `all-mpnet-base-v2` (Reimers & Gurevych, 2019; Sentence-Transformers, 2021b) from the sentence-transformers library, producing 768-dimensional dense vectors. No fine-tuning is applied. Encoder choice is a principal robustness axis: gap rankings are re-computed under two alternative encoders, `all-MiniLM-L6-v2` (Sentence-Transformers, 2021a) (384-dimensional) and `SciBERT` (Beltagy et al., 2019) (768-dimensional). Embeddings were L2-normalised to unit vectors so that cosine similarity equals their dot product.

### 3.4 Token-Aware Segmentation

All source documents are segmented at sentence boundaries using the embedding model’s own tokenizer to enforce a hard token budget, guaranteeing that every segment fits within the encoder’s context window. The procedure applies identically to research abstracts, policy documents, and reference corpora: NLTK sentence-boundary detection is followed by greedy accumulation of sentences up to a token budget of `max_seq_length`  $-10$  (for the canonical MPNet encoder,  $384 - 10 = 374$  tokens). Segments falling below 20 words are discarded; a single sentence exceeding the token budget is kept whole because mid-sentence splitting is not permissible.

For the research-policy comparison this design has two consequences. First, research abstracts (mean  $\sim 200$  words) mostly fit within the 374-token budget, but 17.18% exceed it and split into multiple segments; the research-side text unit is therefore the abstract, whose segments are aggregated to a single document vector before assignment, not the individual segment. Second, policy documents of variable length are split into multiple segments of comparable size to research abstracts, making the two corpora directly comparable in embedding space. The per-document segment cap of  $K = 50$  applied when computing semantic-gap centroids (Section 3.7) is a separate, additional restraint on document-level dominance within a single SDG cluster; it does not affect the segmentation itself.

### 3.5 Supervised Reference Classifier

The canonical instrument is a supervised multinomial logistic regression (LR) classifier trained on pooled single-label reference data from five independent annotated corpora:

- **OSDG Community Dataset** (Pukelis et al., 2022) (30,532 texts, SDGs 1–16): crowdsourced excerpts validated by multiple volunteers.

- **SDG Classification Benchmark** (SDG Classification Expert Group, 2025) (281 expert-verified texts, SDGs 1–17): expert-annotated snippets with consensus resolution.
- **IISD SDG Knowledge Hub** (Wulff et al., 2023) (6,122 texts, SDGs 1–17): journalism and policy commentary on 2030 Agenda implementation.
- **SDGi Corpus** (SDGi, 2024) (20,267 texts, SDGs 1–17): government Voluntary National and Local Reviews.
- **Aurora dataset** (Vanderfeesten et al., 2020) (4,971 expert-validated texts, SDGs 1–17): academic abstracts validated by domain researchers.

OSDG alone contributes about 49% of labelled texts and draws on UN-adjacent documents, so the reference data remain weighted toward policy vocabulary. OSDG and the Benchmark bypass segmentation; the Knowledge Hub, SDGi, and Aurora are segmented before pooling.

The pooled reference data ( $n=62,173$  after single-label filtering) were split into training and held-out test sets ( $n=52,835$  train,  $n=9,338$  test) by stratified document-grouped splitting, keeping all texts from the same source document in one fold to enforce strict separation between training and evaluation. The champion LR ( $C = 3.0$ , L2 penalty, `class_weight=None`, solver `lbfgs`) was selected by 5-fold GroupKFold cross-validation (Hastie et al., 2009) on the training pool and reaches macro- $F_1 = 0.816$  on the held-out test set (Section 4.1). The grid-search protocol, hyperparameter rationale, and the MLP robustness comparison are reported in Appendix G. The validation task does not evaluate a rejection option for non-SDG texts; this is acceptable because the research corpus is first restricted to an OpenAlex AI-and-SDG universe, so the classifier answers a relative question among the 17 SDGs.

Each text is assigned to its highest-probability SDG ( $\text{argmax}$ ), traceable to 768 signed coefficients per class with no hidden layers, no dropout, and no random-seed sensitivity. Macro- $F_1$ , the unweighted mean of per-SDG scores, is the primary validation metric (Section 4.1). This follows the linear-probing paradigm (Alain & Bengio, 2018; Conneau et al., 2018): a linear classifier on frozen embeddings diagnoses representation quality without altering the embedding space.

Because the classifier is trained on pooled reference data that are themselves policy-adjacent, the scoring instrument behaves differently on the two corpora. Policy segments receive systematically higher LR predicted probabilities than research abstracts (mean 0.749 vs 0.627, a 0.122 gap). The gap has two causes the data cannot separate: a genre artefact (the reference data skew toward policy vocabulary) or a substantive difference (policy texts address SDG implementation more directly). Either way the asymmetry does not threaten the findings: hard  $\text{argmax}$  assignment uses only within-corpus ranking, not absolute probability levels, so the directional gap does not reorder goals. It is not an SDGi

circularity effect: the training set and scored policy corpus are disjoint by construction.

### 3.6 Coverage Gap Analysis

For each corpus item, the predicted SDG is the LR argmax (hard assignment, Section 3.5); the coverage profile is the proportion of items assigned to each SDG.

The policy corpus is not a flat set of documents: its 40,597 segments are grouped into 6,367 source documents of very different lengths (Section 3.2). A segment-weighted profile would let a handful of long reports dominate the count, so the analysis instead computes a *document-weighted* profile in which every source document counts equally regardless of length, following the standard centroid formulation in which each document contributes one length-normalised unit vector (Manning et al., 2008). Concretely: segment vectors are averaged per document, hard-assigned to the top SDG, and the coverage profile counts documents (not segments) per SDG. The research corpus uses the same logic at the paper level: each of the 2,536,771 abstracts is represented by the mean of its segments, hard-assigned as a single document unit, so the research-side text unit is the abstract and the two corpora are weighted identically.

The coverage gap for SDG  $j$  is defined as:

$$\text{CoverageGap}_j = \left| \text{ResearchCoverage}_j - \text{PolicyCoverage}_j^{\text{docwt}} \right|$$

The signed version of the same quantity is:

$$\text{SignedCoverageGap}_j = \text{ResearchCoverage}_j - \text{PolicyCoverage}_j^{\text{docwt}}$$

A positive value means research dominates SDG  $j$ ; a negative value means policy dominates.

### 3.7 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG  $j$ , how differently do they discuss it? For each SDG  $j$ , the analysis collects all assigned research papers and all assigned policy segments (capped at 50 per source document to prevent single-document dominance; SDSN reports alone contribute up to 3,179 segments otherwise), computes the L2-normalised mean embedding of each cluster ( $\mathbf{c}_j^{\text{res}}$ ,  $\mathbf{c}_j^{\text{pol}}$ ), and defines:

$$\text{SemanticGap}_j = 1 - (\mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions; a gap approaching 1 indicates

orthogonal framing. The centroid of each cluster is the L2-normalised mean of its segment embeddings; cosine similarity between centroids is the standard method for document-level semantic comparison in embedding space (Babić et al., 2020; vor der Brück & Pouly, 2019). Centroid cosine distance collapses each embedding cloud to its mean direction; Word Mover’s Distance instead solves an optimal-transport problem at the word level (Kusner et al., 2015). Appendix E shows the adjusted-gap ranking correlates moderately with full-distribution alternatives ( $\rho = 0.3\text{--}0.7$ ), confirming the centroid estimate captures the dominant signal while distribution shape carries a related but distinct component.

The *per-document segment cap* of  $K = 50$  (seeded, reproducible) prevents single-document dominance; sensitivity at  $K = 20$  and no cap is reported in Table 13.

### 3.8 Register Adjustment via Iterative Nullspace Projection (INLP)

The semantic gap is not a clean measure of substantive disagreement. Following Biber and Conrad’s distinction between register, genre, and style (Biber & Conrad, 2009), this study treats the research–policy contrast as a composite of register and topic: the two corpora differ in situation of use, audience, and communicative purpose (register), and also in substantive content (topic). Embedding distance therefore mixes genuine within-SDG substantive divergence with register divergence in abstraction, modality, and institutional stance (Biber, 1988). To separate these components, the semantic gap is decomposed into an adjusted semantic gap and a register component using Iterative Nullspace Projection (Ravfogel et al., 2020, INLP). After register removal, the adjusted gap is the primary estimate of within-SDG topic divergence, while the residual register component captures the removed register signal (Table 2).

**Objective.** Following Ravfogel et al. (2020), the goal is to learn a linear projector that removes corpus-identity information from embeddings while preserving distances as much as possible. After  $K$  iterations the composite projector  $P$  removes all linearly decodable corpus signal; the orthogonal nullspace projection is the least harmful linear operation for this purpose.

**SDG-stratified procedure.** Vanilla INLP removes a binary attribute (e.g. gender). The present adaptation must remove *register*, a corpus-level property that is confounded with SDG topic in the raw embeddings. The key modification is SDG stratification: at each iteration, exactly 1000 research and 1000 policy embeddings are drawn from each of the 17 SDGs, preventing the classifier from learning topic membership as a proxy for register. The procedure proceeds as follows at each iteration  $k = 1, \dots, K$ :

1. **Stratified sample.** Draw  $n_{\text{target}} = 1000$  texts per SDG per corpus (global-minimum rule:  $n_{\text{target}} = \min_j \min(|X_j^{\text{res}}|, |X_j^{\text{pol}}|)$ ); under the frozen snapshot,  $n_{\text{target}} = 1207$  for



MPNet). Form the 34-class stratification key  $z_{\text{strat}} = 2 \times \text{sdg\_labels} + y$ , ensuring balanced corpus  $\times$  topic combinations.

2. **Project existing directions.** Subtract from each sample embedding the components along all previously learned directions:  $\tilde{\mathbf{x}} = \mathbf{x} - \sum_{j < k} (\mathbf{g}_j \cdot \mathbf{x}) \mathbf{g}_j$ .
3. **Train classifier.** Fit L2-regularised logistic regression ( $C = 1.0$ , solver: lbfgs, max\_iter: 1000) to predict corpus identity  $y$  from the projected embeddings  $\tilde{\mathbf{x}}$ . Split 15% held-out, stratified by  $z_{\text{strat}}$ .
4. **Extract direction.** Normalise the weight vector:  $\mathbf{g}_k = \mathbf{w}_k / \|\mathbf{w}_k\|$ . Orthogonalise against all prior directions:  $\mathbf{g}_k \leftarrow \mathbf{g}_k - \sum_{j < k} (\mathbf{g}_k \cdot \mathbf{g}_j) \mathbf{g}_j$ . If  $\|\mathbf{g}_k\| < 10^{-8}$  (direction collapse), stop.
5. **Stopping criterion.** Stop if held-out accuracy  $\leq 0.5$  (the majority-class baseline for balanced binary classification). Otherwise append  $\mathbf{g}_k$  and continue.

The INLP projection iterates until held-out accuracy falls to  $\leq 0.5$  (majority-class baseline). The implementation imposes a safety cap that is never reached (all three encoder tracks converge within 40–79 iterations; Appendix H.1). The final adjusted embedding is  $\mathbf{x}' = \mathbf{x} - \sum_{k=1}^K (\mathbf{g}_k \cdot \mathbf{x}) \mathbf{g}_k$ , and the adjusted gap uses centroids computed on these projected embeddings. Convergence behaviour is reported in Appendix H.1 (Table 10).

**Identification argument.** The stratification design is the critical structural precondition: because every iteration trains on equal numbers of research and policy embeddings from each SDG, no linear direction can simultaneously encode SDG topic and corpus identity. What remains linearly decodable is the corpus-level register difference: modality, audience conventions, institutional stance, and structural patterns. The removed subspace is interpreted as register in the sense of Biber and Conrad (2009): a variety associated with a particular situation of use, including communicative purpose, audience, and production circumstances. Because stratification balances SDG topic but not other variety axes, the removed component may combine register with period effects and author-background signals, so the register reading is a supported interpretation rather than an exhaustive characterisation (Grieve et al., 2025). This interpretation is evaluated against independent linguistic markers in Appendix H.3; the validation finds substantial reduction of the corpus-level signal but not its complete elimination.

With the projector learned, the semantic-gap analysis of Section 3.7 is recomputed twice: once on the adjusted embeddings, yielding the within-SDG *adjusted gap* (the primary estimate of topic divergence), and once on the removed component  $(\mathbf{x} - \mathbf{x}')$ , yielding the *register component* (the amount of the semantic gap attributable to register).

### 3.9 Coverage–Semantic Interaction

Their operationalization is as follows. Pearson  $r$  and Spearman rank correlation  $\rho$  (Spearman, 1904) are computed between each coverage predictor (coverage gap, research–policy dominance, research coverage, policy coverage) and the within-SDG semantic-gap scores across the 17 SDGs; SDG 4 is treated as artefact-affected and examined separately (Appendix C.1).

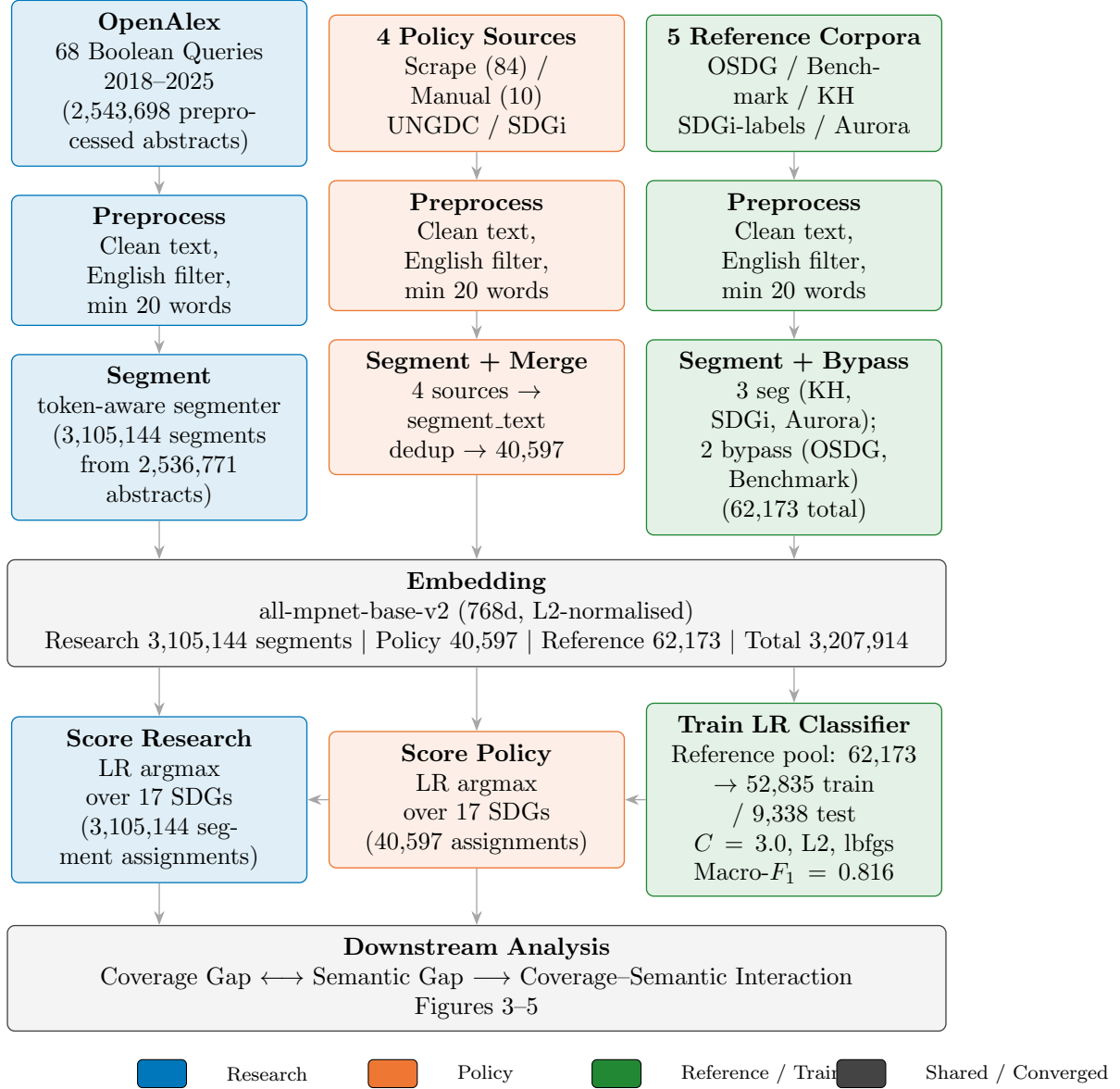
The 17 SDGs are the complete framework, not a sample; the observed associations are properties of the framework, and the only stochastic element is the measurement process (classifier assignment, corpus construction, embeddings). All reported  $p$ -values are two-sided Monte Carlo permutation  $p$ -values (100,000 resamples, fixed seed): they condition on the measured values and test the null of no association between them, so no sampling-based power reasoning is involved (Abadie et al., 2020; Berk et al., 1995).

The operative small- $n$  concern is robustness rather than power: with only 17 framework units, individual goals can be influential, so the associations are read together with the leave-one-out check (Section 4.3) and the cross-configuration checks (Section 4.4).

The raw, adjusted, and register-component correlations are a single decomposition (register = raw – adjusted), mechanically linked; the substantive claim concerns their joint cancellation pattern, not three separate significance decisions. All three are reported unconditionally for every configuration, so the family is fully disclosed. Because the decomposition is pre-specified (Section 3.8) and the statistics are jointly informative, no familywise correction is applied.

Finer-grained units (the framework defines 169 targets) would raise the observation count, but no labelled target-level reference data exist. A supplementary pooled-regression analysis combines all 24 configurations into a single panel (Appendix J).

Figure 2 summarises the end-to-end methodology.



**Figure 2.** End-to-end methodology pipeline in three parallel data tracks.

## 4 Results

This section reports validation, coverage gaps, semantic gaps, H1 tests, and robustness without interpretation.

### 4.1 Supervised Reference Classifier and Coverage Gap

The LR classifier was evaluated on the held-out test set (n=9,338, document-grouped to prevent source leakage). It reaches macro- $F_1 = 0.816$  and micro- $F_1 = 0.8306$ , far above the 1/17 random baseline (0.059). Per-class F1 values from the test set are reported in Table 1.

**Table 1.** LR test-set F1 scores (n=9,338).

| SDG                           | LR test F1   |
|-------------------------------|--------------|
| SDG 1 (No Poverty)            | 0.737        |
| SDG 2 (Zero Hunger)           | 0.830        |
| SDG 3 (Good Health)           | 0.903        |
| SDG 4 (Quality Education)     | 0.889        |
| SDG 5 (Gender Equality)       | 0.881        |
| SDG 6 (Clean Water)           | 0.879        |
| SDG 7 (Clean Energy)          | 0.861        |
| SDG 8 (Decent Work)           | 0.716        |
| SDG 9 (Industry & Infra.)     | 0.770        |
| SDG 10 (Reduced Inequalities) | 0.610        |
| SDG 11 (Sustainable Cities)   | 0.762        |
| SDG 12 (Responsible Cons.)    | 0.783        |
| SDG 13 (Climate Action)       | 0.831        |
| SDG 14 (Life Below Water)     | 0.892        |
| SDG 15 (Life on Land)         | 0.865        |
| SDG 16 (Peace & Justice)      | 0.879        |
| SDG 17 (Partnerships)         | 0.788        |
| Macro-F1 (SDGs 1–17)          | <b>0.816</b> |

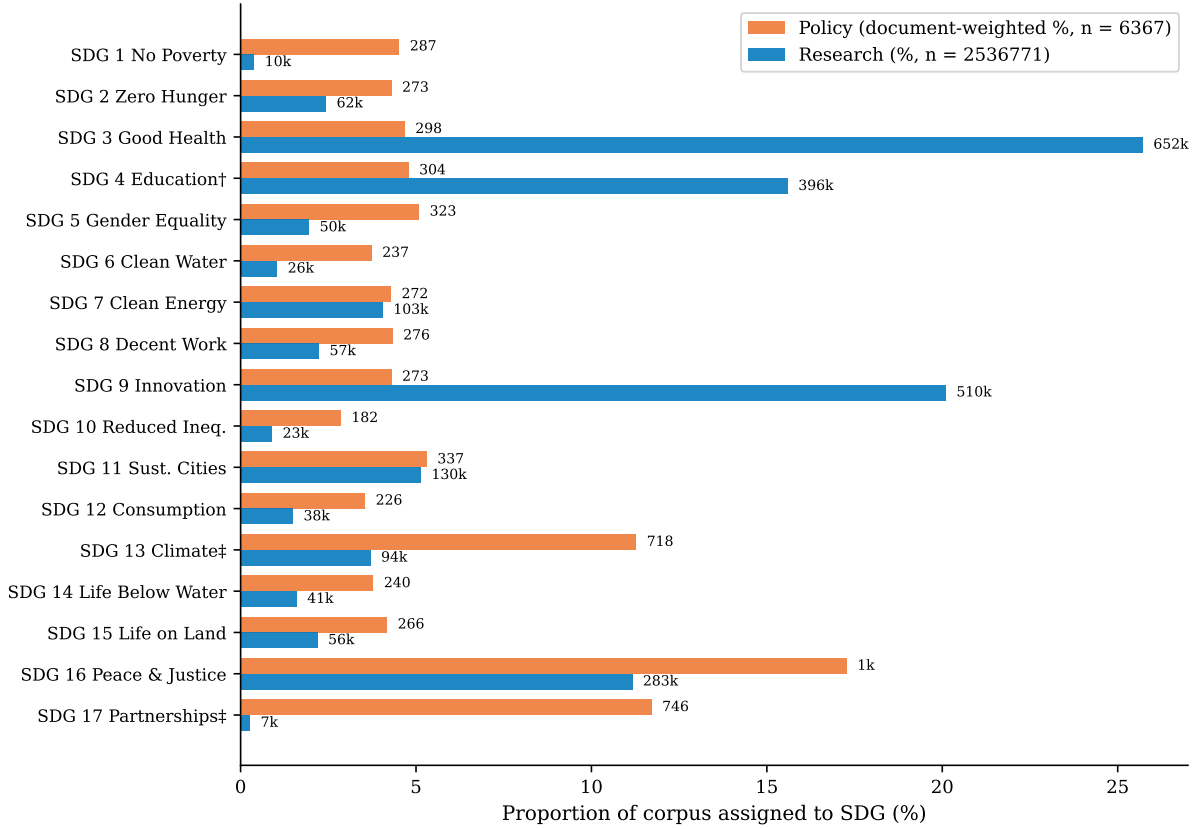
*Notes:* LR test F1 on the held-out pooled test set (all reference sources). The table shows the canonical 17-way classifier used throughout the main analysis.

Per-class F1 ranges from 0.610 (SDG 10) to 0.903 (SDG 3); most goals score between 0.76 and 0.89. The weakest classes (SDGs 10 and 8) are consistent with their semantic breadth (Kashnitsky et al., 2024) (Appendix C.2, Figure 7).

The MLP robustness check achieves comparable performance (MLP: 0.826; LR: 0.816), confirming architecture choice is not the binding constraint (Appendix G).

Figure 3 presents the document-weighted policy coverage profile alongside the research coverage profile.

**Policy corpus.** Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 16 (Peace, Justice and Strong Institutions, 17.3%), SDG 13 (Climate Action, 11.3%), and SDG 17 (Partnerships for the Goals, 11.7%), accounting



**Figure 3.** Coverage profiles by SDG. SDG 4 is flagged as artefact-affected due to ML-learning vocabulary overlap with education corpora (Appendix C.1).

for 40.3% of policy documents. This ordering is consistent with the literature review’s account of SDG 17 as the cross-cutting means-of-implementation goal (Section 2) (Griggs et al., 2017; Le Blanc, 2015). The SDG 16 lead is notable: under the supervised classifier, governance and justice discourse emerges as the most prevalent policy theme, ahead of both climate and partnerships. Appendix I.6 shows this full-corpus profile is source-family sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, while SDGi and UNGDC are more climate-, partnerships-, and governance-oriented.

**Research corpus.** The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 25.7%), SDG 9 (Innovation, 20.1%), and SDG 4 (Quality Education, 15.6%), followed by SDG 16 and SDG 11 at 11.2% and 5.1% respectively. The SDG 3 lead corroborates the literature review, where SDG 3 is the one goal every major AI and sustainability bibliometric study finds dominant Bautista-Puig et al. (2021), Cowls et al. (2021), Nedungadi et al. (2024), and Singh et al. (2024), and Gohr et al. likewise show health among the largest SDG-focus shares in their reviewed corpus (Gohr et al., 2025, Fig. 1). The SDG 4 share is the one point of disagreement: prior inventories do not rank SDG 4 among the leaders, so its third-place finish here is the same result flagged as likely inflated by machine-learning vocabulary (Appendix C.1). Even so, the research corpus clearly does not mirror the

policy-side dominance of SDGs 16, 13, and 17; the research-side near-absence of SDG 17 (0.3%) and SDG 1 (0.4%) matches Gohr et al.’s finding that SDG 1, SDG 5, and SDG 17 are severely underrepresented in the AI–SDG literature (Gohr et al., 2025).

**Largest coverage mismatches.** The five largest absolute coverage gaps are SDG 3 (0.210), SDG 9 (0.158), SDG 17 (0.114), SDG 4<sup>1</sup> (0.108), and SDG 13 (0.076). The largest single-goal imbalance is SDG 3 (research 25.7% vs. policy 4.7%): research concentrates on technically tractable health applications while policy underweights the goal. The most extreme policy-over-research imbalances are SDG 17 (policy 11.7% vs. research 0.3%) and SDG 13 (11.3% vs. 3.7%): policy frames partnerships and climate as cross-cutting coordination challenges, while research engages them far less. SDG 16 is policy’s top theme (17.3%) but research does engage it (11.2%, third-highest research share), so it is not a “barely-touches” case. The SDG 17 mismatch is corroborated independently: Gohr et al. found no SDG 17 articles at all among the 792 AI–SDG papers they reviewed (Gohr et al., 2025).

The coverage-gap ranking is stable across embedding architectures and retrieval strategy; the cross-encoder, cross-retrieval, cross-segment-cap, and cross-policy-source-family rankings are reported in Appendix I (Tables 18, 17, 13).

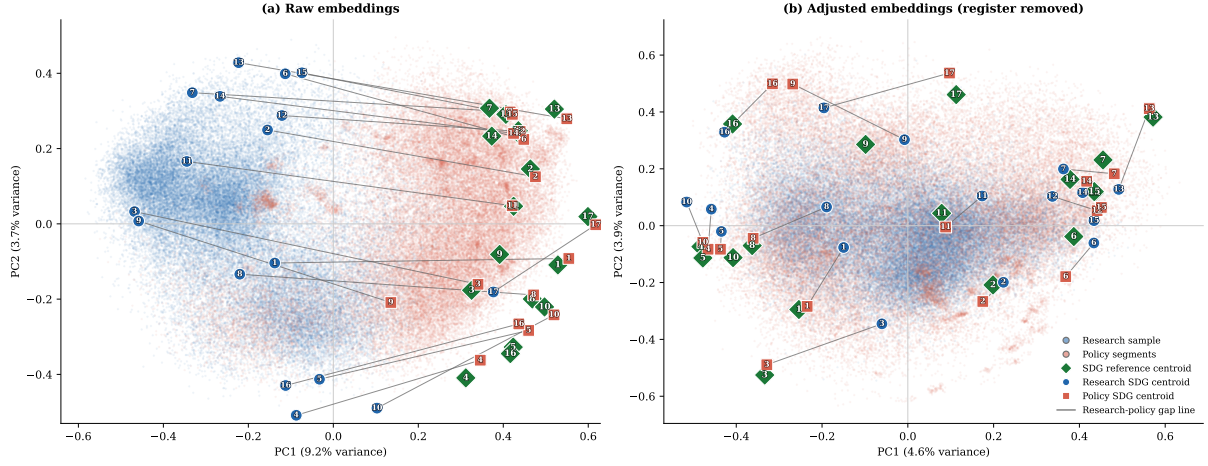
## 4.2 Semantic Gap after Register Removal

Figure 4 shows the combined embedding space and the geometric effect of INLP register removal: the raw space (left) separates research and policy clouds, while the adjusted space (right) merges them, confirming that the separation is driven by register rather than topic. The policy cloud hugs the 17 reference centroids more tightly than the research cloud. The projection is descriptive only (9.2% and 3.7% of variance); cosine distances remain in the original 768-dimensional space. The effect is encoder-dependent, since SciBERT already fuses the clouds without INLP, so this figure is MPNet-specific.

Figure 5 shows the within-SDG semantic gap for all 17 SDGs, sorted by the adjusted (topic) gap after register removal. The adjusted gap ranges from 0.326 (SDG 17, widest) to 0.084 (SDG 15, narrowest), a span of 0.241 cosine units. The raw gap (naive baseline) ranges from 0.494 (SDG 3, widest) to 0.177 (SDG 17, narrowest), a span of 0.317. The adjusted ranking is stable when the segment cap is 20 per document or removed (Table 13), so the result does not depend on the sampling parameter. A second sampling concern is corpus asymmetry: research (2,536,771 abstracts) outweighs policy (6,367 documents) by roughly 398:1 in document units. This study therefore re-derived the ranking on random research draws comparable in size to the policy corpus: across 100 independent 50,000-paper draws it reproduces the full-corpus ordering at Spearman  $\rho = 0.986$  (SD

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<sup>1</sup>SDG 4’s share is partly inflated by machine-learning vocabulary overlap (Appendix C.1).



**Figure 4.** PCA before/after INLP register removal (MPNet).

0.009), so the research-side over-representation does not drive which SDGs show large versus small gaps (Appendix F).

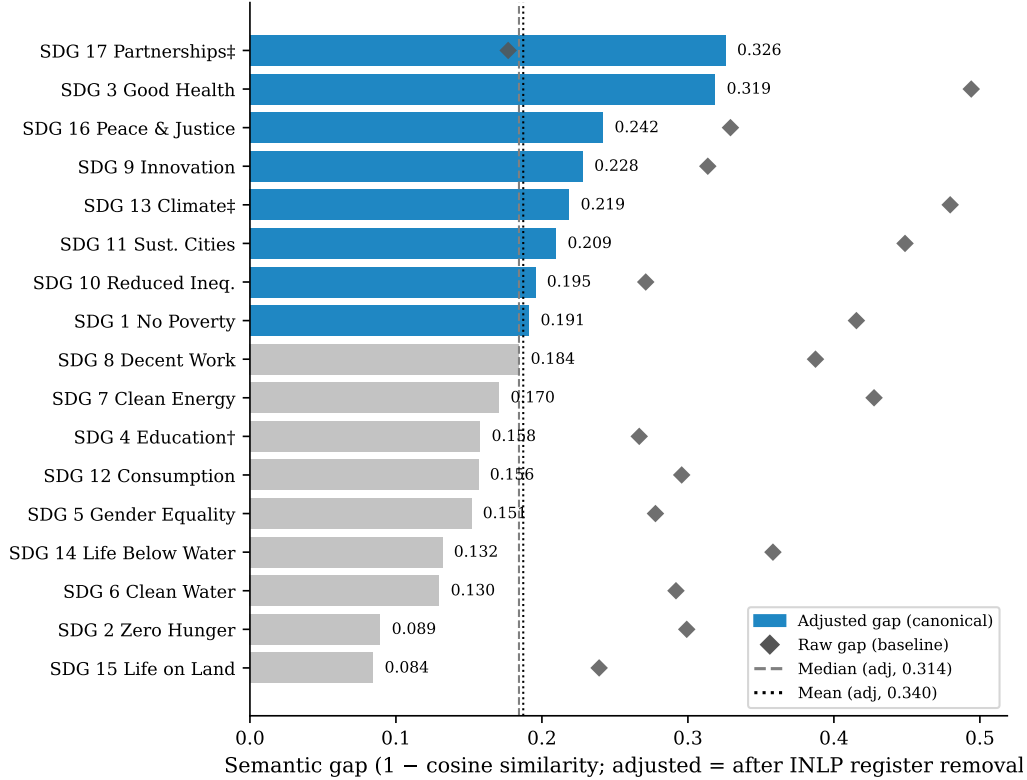
**Largest semantic gaps (adjusted, primary).** SDGs 17, 3, 16, 9, and 13 show the widest adjusted gaps. Under the raw gap, the ordering shifts: SDG 3 (0.494) leads. The register decomposition confirms that register uniformly inflates the raw gap, except for SDG 17 where register *increases* apparent similarity.

**Table 2.** Register-topic decomposition of the semantic gap.

| SDG    | Description                             | Semantic Gap | Adj. Gap | Reg. Comp. | Cov. Gap |
|--------|-----------------------------------------|--------------|----------|------------|----------|
| SDG 1  | No Poverty                              | 0.415        | 0.191    | +0.224     | 0.0413   |
| SDG 2  | Zero Hunger                             | 0.299        | 0.089    | +0.210     | 0.0187   |
| SDG 3  | Good Health and Well-Being              | 0.494        | 0.319    | +0.176     | 0.2104   |
| SDG 4  | Quality Education                       | 0.267        | 0.158    | +0.109     | 0.1081   |
| SDG 5  | Gender Equality                         | 0.278        | 0.151    | +0.126     | 0.0314   |
| SDG 6  | Clean Water and Sanitation              | 0.292        | 0.130    | +0.162     | 0.0269   |
| SDG 7  | Affordable and Clean Energy             | 0.428        | 0.170    | +0.257     | 0.0023   |
| SDG 8  | Decent Work and Economic Growth         | 0.387        | 0.184    | +0.203     | 0.0210   |
| SDG 9  | Industry, Innovation and Infrastructure | 0.314        | 0.228    | +0.086     | 0.1580   |
| SDG 10 | Reduced Inequalities                    | 0.271        | 0.195    | +0.076     | 0.0196   |
| SDG 11 | Sustainable Cities and Communities      | 0.449        | 0.209    | +0.239     | 0.0017   |
| SDG 12 | Responsible Consumption and Production  | 0.296        | 0.156    | +0.139     | 0.0206   |
| SDG 13 | Climate Action                          | 0.480        | 0.219    | +0.261     | 0.0757   |
| SDG 14 | Life Below Water                        | 0.358        | 0.132    | +0.227     | 0.0219   |
| SDG 15 | Life on Land                            | 0.239        | 0.084    | +0.155     | 0.0198   |
| SDG 16 | Peace, Justice and Strong Institutions  | 0.329        | 0.242    | +0.087     | 0.0611   |
| SDG 17 | Partnerships for the Goals              | 0.177        | 0.326    | -0.149     | 0.1145   |

*Notes:* Raw gap is the reference semantic-gap estimate; Adjusted gap is after INLP register removal (primary estimate; evaluated in Appendix H.3); Register component = raw – adjusted. Coverage correlations are Spearman  $\rho$  across 17 SDGs.

**Smallest semantic gaps (adjusted, primary).** SDG 15 (Life on Land, 0.084), SDG 2



**Figure 5.** Within-SDG semantic gap by SDG, sorted by the adjusted (topic) gap after register removal.

(Zero Hunger, 0.089), SDG 6 (Clean Water, 0.130), SDG 14 (Life Below Water, 0.132), and SDG 5 (Gender Equality, 0.151) show the narrowest adjusted gaps. A narrow gap does not mean balanced attention: SDG 9 is still research-dominant, and SDG 4 carries the learning-vocabulary concern (Appendix C.1). Appendix D shows example texts for high- and low-gap cases.

The semantic-gap ranking is stable across encoders, retrieval strategy, segment cap, and policy-source family (Appendix I).

### 4.3 Coverage–Semantic Interaction: Results

Table 3 reports H1a–H1d rank correlations between coverage predictors and the within-SDG semantic gap across all encoder–classifier configs. Columns show Spearman  $\rho$  against the raw gap, adjusted gap, and register component. Unlike the rank-robustness tables, the three columns are adjacent because the apparent cancellation *for each config* is the object of comparison.

Predictors are research coverage, policy coverage, absolute coverage gap, and signed dominance. MLP configs reuse their encoder’s INLP matrix; Concept rows use the concept-retrieved research corpus (Appendix B).



**Table 3.** H1a–H1d coverage-predictor vs semantic-gap rank correlations (Spearman  $\rho$ ) across encoder–classifier configs.

|                                                                                 | Semantic Gap        | Adj. gap            | Reg. Comp.          |
|---------------------------------------------------------------------------------|---------------------|---------------------|---------------------|
| <b>H1a. Coverage gap <math>\leftrightarrow</math> Semantic gap</b>              |                     |                     |                     |
| MPNet LR                                                                        | -0.012              | +0.544*             | -0.390              |
| MPNet MLP                                                                       | -0.135              | +0.336              | -0.370              |
| MPNet ZS                                                                        | -0.299              | +0.203              | -0.449 <sup>†</sup> |
| MiniLM LR                                                                       | -0.243              | +0.397              | -0.463 <sup>†</sup> |
| MiniLM MLP                                                                      | -0.042              | +0.505*             | -0.272              |
| SciBERT LR                                                                      | -0.130              | +0.221              | -0.267              |
| SciBERT MLP                                                                     | -0.333              | -0.034              | -0.120              |
| Concept LR                                                                      | +0.108              | +0.434 <sup>†</sup> | -0.194              |
| Concept MLP                                                                     | -0.017              | +0.181              | -0.140              |
| <b>H1b. Policy–research dominance <math>\leftrightarrow</math> Semantic gap</b> |                     |                     |                     |
| MPNet LR                                                                        | +0.159              | -0.059              | +0.029              |
| MPNet MLP                                                                       | +0.020              | -0.118              | -0.034              |
| MPNet ZS                                                                        | -0.005              | -0.336              | +0.360              |
| MiniLM LR                                                                       | +0.404              | -0.020              | +0.429 <sup>†</sup> |
| MiniLM MLP                                                                      | +0.319              | -0.066              | +0.348              |
| SciBERT LR                                                                      | +0.549*             | -0.118              | +0.547*             |
| SciBERT MLP                                                                     | +0.336              | -0.429 <sup>†</sup> | +0.529*             |
| Concept LR                                                                      | +0.051              | -0.029              | -0.260              |
| Concept MLP                                                                     | -0.174              | +0.093              | -0.199              |
| <b>H1c. Research coverage <math>\leftrightarrow</math> Semantic gap</b>         |                     |                     |                     |
| MPNet LR                                                                        | +0.449 <sup>†</sup> | +0.267              | +0.145              |
| MPNet MLP                                                                       | +0.262              | +0.098              | +0.088              |
| MPNet ZS                                                                        | -0.100              | -0.328              | +0.314              |
| MiniLM LR                                                                       | +0.444 <sup>†</sup> | +0.262              | +0.331              |
| MiniLM MLP                                                                      | +0.373              | +0.275              | +0.279              |
| SciBERT LR                                                                      | +0.605*             | +0.466 <sup>†</sup> | +0.020              |
| SciBERT MLP                                                                     | -0.005              | -0.196              | +0.225              |
| Concept LR                                                                      | +0.221              | +0.017              | -0.076              |
| Concept MLP                                                                     | -0.098              | -0.010              | -0.127              |
| <b>H1d. Policy coverage <math>\leftrightarrow</math> Semantic gap</b>           |                     |                     |                     |
| MPNet LR                                                                        | +0.212              | +0.589*             | -0.009              |
| MPNet MLP                                                                       | +0.175              | +0.428 <sup>†</sup> | -0.007              |
| MPNet ZS                                                                        | -0.094              | +0.240              | -0.321              |
| MiniLM LR                                                                       | -0.015              | +0.449 <sup>†</sup> | -0.159              |
| MiniLM MLP                                                                      | +0.083              | +0.456 <sup>†</sup> | -0.093              |
| SciBERT LR                                                                      | -0.100              | +0.605*             | -0.686**            |
| SciBERT MLP                                                                     | -0.172              | +0.395              | -0.395              |
| Concept LR                                                                      | +0.369              | +0.510*             | +0.178              |
| Concept MLP                                                                     | +0.256              | +0.258              | +0.082              |

*Notes:* Raw gap = main semantic-gap estimate; Adj. gap = after INLP register removal (main reported estimate; evaluated in Appendix H.3); Register = raw – adjusted. Each correlation is between two rank vectors (ranks 1–17 across the 17 SDGs); Spearman  $\rho$  is the Pearson correlation of these ranks, testing monotonic association. \*\*\* $p < .001$ , \*\* $p < .01$ , \* $p < .05$ , <sup>†</sup> $p < .10$ .

H1a is null on the raw gap ( $\rho = -0.012$ ,  $p = 0.959$ ; Table 3, MPNet LR). The raw gap masks a register–topic decomposition: the register component ( $\rho = -0.390$ ) cancels the topic signal ( $\rho = 0.544$ ). After register removal, the adjusted gap is positively associated with the coverage gap (positive in 8/9 configs). The directional decomposition reveals an asymmetric pattern: H1d (policy coverage) is positive in 9/9 configurations, significant at  $p < 0.05$  for MPNet LR ( $\rho = +0.589$ ) and Concept LR ( $\rho = +0.510$ ), while H1b (signed dominance) and H1c (research coverage) show no consistent adjusted-direction signal. The linkage between coverage and framing operates through the policy side: where policy heavily covers an SDG, topic divergence is wider.

The adjusted association is marginal: under a conservative familywise view of the three linked statistics,  $p = 0.025$  would not survive a correction, so it is read as pattern-consistent evidence for the cancellation rather than a confirmatory test, with the weight carried by the cross-configuration and cross-encoder consistency above (Section 4.4).

A metric-sensitivity check repeats this grid with Pearson  $r$  on the raw magnitudes instead of Spearman  $\rho$  (Appendix I.5); the adjusted-gap association is unchanged for the supervised tracks across all four H1 hypotheses, and the only reversals are the already-flagged lower-trust SciBERT configs, confirming the rank-based reporting is not driving the result.

The SDG 4 lexical artefact (Appendix C.1) does not drive this result. A leave-one-out check with SDG 4 removed leaves the raw coverage-gap correlation null ( $\rho = 0.088$ ,  $p = 0.744$ ) and preserves the positive register-adjusted coverage-gap association ( $\rho = 0.597$ ,  $p = 0.016$ ), so the artefact props up neither the raw null nor the adjusted association.

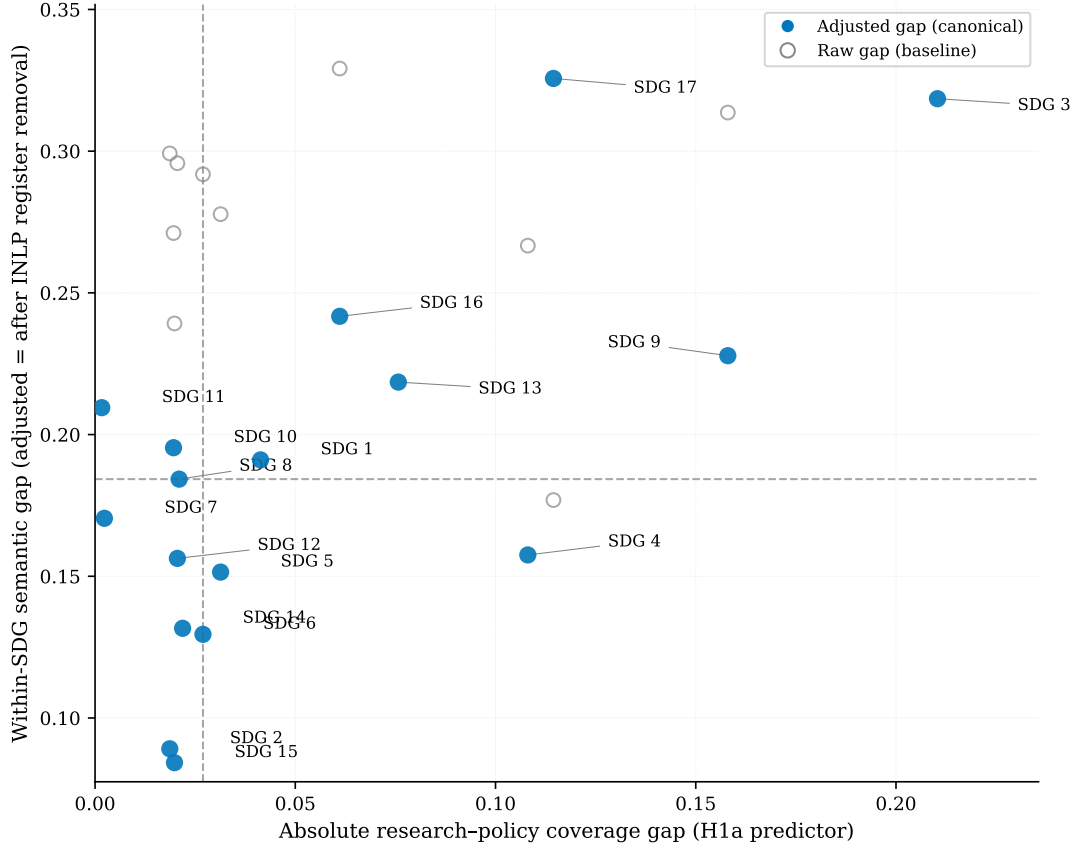
The corpus asymmetry does not manufacture this null. Because the research corpus is the  $\text{AI} \cap \text{SDG}$  intersection while the policy corpus pools AI-specific and broad institutional SDG discourse, this study re-ran the interaction test with the policy side restricted to the curated AI/SDG family, a symmetric AI/SDG-vs-AI/SDG comparison that matches the research corpus’s scope. The cancellation pattern holds: research share remains uncorrelated with the gap ( $\rho = -0.06$ ,  $p = 0.833$ ), and the coverage-gap correlation does not reach significance either ( $\rho = -0.22$ ,  $p = 0.386$ ; Table 24, Appendix I.6).

Figure 6 shows adjusted semantic gap vs. coverage gap; open markers show the raw baseline. Large coverage gaps span the full range of adjusted semantic gaps; no single SDG drives the weak H1a association.

## 4.4 Robustness of the Gap Rankings

Register removal is validated against independent linguistic markers (Appendix H.3); the adjusted gap is the primary estimate.

**Cross-sensitivity (policy source, segment cap, retrieval).** Under MPNet–LR, the



**Figure 6.** Coverage gap vs semantic gap, one point per SDG.

semantic-gap ranking is stable across policy source, segment cap, and retrieval ( $\rho \geq 0.65$ ; Table 13). The same holds for coverage (Table 18).

**Encoder sensitivity.** Coverage-gap ranks are stable across encoders (MPNet-LR vs. SciBERT-LR  $\rho = 0.93$ ; vs. MiniLM-LR  $\rho = 0.89$ ; Table 17). Semantic-gap magnitude is encoder-dependent (SciBERT: 0.04 vs. MPNet: 0.34) because scientific pretraining fuses the genres (Beltagy et al., 2019), but the divergence conclusion holds (Table 15). Under the raw gap, SDG 3 leads for MPNet-LR while SDG 13 leads for MiniLM-LR and SciBERT-LR; SDG 17 trails under all three.

**Assignment-method sensitivity.** LR and MLP show varied agreement (MPNet LR vs. MLP  $\rho = 0.85$ ; Table 15). SDG 17 is most method-sensitive (rank 17 under LR, rank 8 under zero-shot).

**Distance-functional robustness.** Distribution-aware metrics (sliced Wasserstein, Chamfer, Gaussian-2-Wasserstein, MMD, energy, C2ST) correlate only moderately with the adjusted semantic gap ( $\rho = 0.525$  for sliced Wasserstein; Appendix E). The adjusted gap is primarily a mean-direction finding; shape carries a related but distinct signal.

**Synthesis.** Two conclusions survive: SDG 15 is robustly least-divergent (rank 17 across configs); SDG 17 is most method-sensitive and most divergent under the adjusted gap,

while SDG 3 leads under the raw gap for MPNet-LR. The cancellation replicates across all configs and policy-source families. Pooled OLS confirms the adjusted association (Appendix J).

## 5 Discussion

The headline result is a cancellation: the raw correlation is null because register and topic offset ( $\rho = 0.544$  vs.  $\rho = -0.390$ ; Table 3).

### 5.1 The cancellation is real, not artefactual

Three potential confounds must be ruled out before the cancellation can be interpreted substantively.

**Independence of classification and measurement.** The LR classifier and the embedding space are separate instruments: classification uses softmax probabilities; measurement uses cosine distance between L2-normalised centroids.

**Cross-configuration replication.** The cancellation replicates across 8 of 9 configs (Table 3), including concept retrieval and MLP; the register component is negative in 7 of 9; the raw gap is null in all.

**SciBERT compression validates the register interpretation.** SciBERT compresses the raw gap from 0.34 to 0.04 (88% reduction) while preserving coverage ( $\rho = 0.93$ ; Table 17). Scientific pretraining (Beltagy et al., 2019) fuses the registers because both use scientific conventions, exactly what the decomposition predicts.

### 5.2 Two communities, two dimensions

If the cancellation is real, it demands a theoretical explanation. The two-communities theory (Caplan, 1979) provides one. Caplan characterises social scientists and policy makers as living in “separate worlds with different and often conflicting values, different reward systems, and different languages” (Caplan, 1979, p. 459). The gap, he argues, “necessarily involves value and ideological dimensions as well as technical ones” (Caplan, 1979, p. 461). If the two communities are genuinely distinct knowledge cultures, then a near-zero raw correlation is *expected*, not puzzling: the two corpora develop different topical priorities (producing coverage divergence) and different communicative norms (producing register divergence), and the two effects operate in opposite directions on the semantic gap. This is not a communication failure. Caplan himself warned that “it does not follow... that an alliance of social scientists and policy makers is the panacea” (Caplan, 1979, p. 461). The raw null is the expected signature of two distinct knowledge cultures

coexisting without deep integration.

The adjusted association reveals something further: coverage and framing are *separable dimensions*, not a single “alignment” property. As shown in Results (Section 4.1), SDGs 9 and 17 illustrate this: SDG 9 has the second-largest coverage gap but a moderate semantic gap, while SDG 17 has moderate coverage but the widest adjusted gap. A single alignment score would conflate these two genuinely different patterns. This separability is the paper’s core empirical contribution to the SDG-mapping literature, which has treated coverage as the primary output (Armitage et al., 2020; Kashnitsky et al., 2024). The framework demonstrates that coverage and framing should be reported as independent axes.

The positive adjusted correlation ( $\rho = +0.544$ ) further confirms that the two dimensions are linked through topic: where one community dominates a given SDG’s coverage, the framing tends to diverge. The epistemic-communities framework (Haas, 1992) sharpens this reading. Haas argues that “before states can agree on whether and how to deal collectively with a specific problem, they must reach some consensus about the nature and scope of the problem. . .” (Haas, 1992, p. 29). Where research dominates coverage (SDGs 3, 9), the framing reflects academic norms of knowledge production: “problems set and solved in a context governed by the, largely academic, interests of a specific community” (Gibbons et al., 1994, p. 3). Where policy dominates (SDGs 13, 16, 17), the framing reflects institutional norms of coordination: “knowledge carried out in a context of application” (Gibbons et al., 1994, p. 3). The coverage-semantic linkage is the empirical signature of two communities producing different framings of the same goals.

However, Haas also cautions that “given the many examples of different states reacting in different ways to the same consensual evidence provided by specialists, it is unclear how effective consensual knowledge is, as an independent variable, at explaining or predicting state behavior” (Haas, 1992, p. 30). The adjusted association shows that coverage and framing are linked through topic; it does not show that this linkage produces policy uptake.

The directional decomposition (H1b–H1d) reveals an asymmetric structure, as shown in Table 3. H1d (policy coverage) is the strongest consistent predictor: positive in 9 of 9 configurations, significant at  $p < 0.05$  for MPNet LR ( $\rho = +0.589$ ) and Concept LR ( $\rho = +0.510$ ). The linkage operates through the policy side: where policy heavily covers an SDG, framing divergence is wider, consistent with Haas’s observation that epistemic communities occupy institutional niches (Haas, 1992, p. 30) and with the Mode 1/Mode 2 distinction (Gibbons et al., 1994).

This asymmetry is consistent with Cash et al.’s (Cash et al., 2003) observation that salience, credibility, and legitimacy are “tightly coupled” such that efforts to enhance one “normally incur a cost to the others.” Where policy attention is concentrated (SDGs 13, 16, 17), the institutional framing that drives salience simultaneously distances the discourse from

research norms of credibility, widening the adjusted gap. The mechanism is structural, not communicative: Mode 2 knowledge production (Gibbons et al., 1994) shapes policy discourse in contexts of application, while Mode 1 knowledge production shapes research discourse in discipline-driven contexts; where policy dominates an SDG, the Mode 2 framing prevails and the gap widens. This is not a failure of communication between the two communities but the expected signature of two distinct knowledge cultures producing different framings of the same goals.

### 5.3 The register decomposition: what the raw gap actually measures

For 16 of 17 SDGs, the register component is positive (+0.076 to +0.261; Table 2), meaning register uniformly inflates the apparent divergence. The implication is stark: any SDG-mapping study that uses raw embedding distances without register adjustment will overestimate semantic divergence. The raw gap is a composite of topic, register, and convention; only the adjusted gap isolates topic.

The exception is SDG 17, where the register component is negative (-0.149): register makes the two communities appear *more similar* than their topics warrant. Under the raw gap SDG 17 ranks near the bottom (0.177); after register removal it jumps to the top (0.326). Policy partnership discourse uses institutional language (*countries, nations, global*) that superficially resembles research vocabulary, but the substance differs: research concerns technical terms (*model, method*), while policy concerns institutional coordination.

SDG 17 is where framing power operates most invisibly: partnership language *sounds* like either community, but the coordination logic belongs exclusively to policy. LeBlanc (Le Blanc, 2015) and ICSU (Griggs et al., 2017) describe SDG 17 as “implementation infrastructure rather than substantive domains”; the register inversion confirms this empirically.

The register decomposition also reveals what the embedding instrument actually measures. Embedding distance is a composite signal of topic and register (Harris, 1954). The INLP procedure with SDG stratification (Ravfogel et al., 2020) exploits the structural fact that the 17 SDGs supply a near-orthogonal topic set that cross-cuts both registers, allowing linear separation. The method’s applicability is bounded by the ontology: in domains where topic and register are tightly confounded (e.g., clinical notes versus biomedical literature), the projection would leak topic into the register direction.

## 5.4 Implications: topical mention is not alignment

Coverage, the Level 1 metric existing studies report, captures only a weak form of salience; it does not address credibility or legitimacy (Cash et al., 2003). The adjusted gap is a Level 2 metric that begins to measure framing compatibility.

The high-gap SDGs (17, 3, 16, 9, 13) are candidates for boundary-organising practices (Guston, 2001): policy synthesis briefs, joint workshops, or funder-required summaries. Cash et al. (Cash et al., 2003) caution that translation alone cannot resolve the tradeoffs among salience, credibility, and legitimacy.

Existing SDG-mapping studies report only coverage (Armitage et al., 2020); this paper shows coverage and framing are separable dimensions.

## 5.5 Limitations

**Semantic proximity, not substantive alignment.** The study is positioned at Level 2 of a three-level construct (§2.2). Level 3 requires qualitative, institutional, and deliberative methods and is left to future work.

**Distance metric discards distributional shape.** Centroids are mean-direction measures. Under raw embeddings, distribution-aware metrics (sliced Wasserstein, Chamfer) correlated strongly with the semantic gap ( $\rho = 0.44\text{--}0.91$ ), but this agreement was largely a register artifact: the distribution-aware metrics track the register component of the gap, not the adjusted semantic gap (Appendix E). After register removal, the adjusted semantic gap is a mean-direction measure, and distribution-aware metrics correlate only moderately with it (e.g. Gaussian-2-Wasserstein  $\rho = 0.243$ ); they remain an informative full-distribution complement, not a re-ranking. The Wasserstein shape term accounts for  $\approx 79\%$  of the distance. Sampling stability is verified with a replicate seed (maximum gap change 0.008).

**Corpus scope and asymmetric construction.** The research corpus is  $\text{AI} \cap \text{SDG}$  (tight intersection); the policy corpus is broader institutional SDG discourse. The comparison is inherently asymmetric because “AI and the SDGs” is not yet a mainstream organising category in policy discourse. The curated AI/SDG policy family (94 documents) is underpowered for the symmetric check. The asymmetry was addressed by re-testing the central interaction on the curated AI/SDG family (Section 4.3, Appendix I.6); broadening the research side to all-SDG research would complete the symmetry and is left to future work.

**Retrieval-method dependency.** The keyword-retrieved corpus trades recall for precision; Appendix B shows that concept-retrieved papers (field-of-study membership) produce closely aligned gap rankings but avoid the SDG 4 artefact (Appendix C.1). Re-running

the full pipeline with concept-retrieved papers as the primary corpus is the most direct path to eliminating this artefact and is left for future work.

**Register adjustment is partial.** INLP removes a linear subspace that captures the dominant corpus-level register difference, but register is multi-layered (Biber, 1988). The register interpretation rests on a structural argument (SDG stratification ensures topic cannot leak into the removed direction; §3.8) plus partial empirical validation against independent linguistic markers (Appendix H.3). The validation finds substantial reduction of the corpus-level signal but not its complete elimination.

**Per-SDG assignment is approximate.** The hard argmax treats SDG discourse as cleanly separable, but the PCA landscape (Figure 4) shows real-world SDG discourse is fuzzy. SDG 4 is artefact-affected by the machine-learning/education “learning” overlap; its rank is read conditional on the lexical audit (Appendix C.1). Leave-one-out of SDG 4 preserves the adjusted association ( $\rho = 0.597$ ,  $p = 0.016$ ).

**Cross-sectional design.** The study observes one period (2018–2025) against a similarly timed policy corpus; it cannot assess whether alignment is improving or deteriorating, or whether specific events (e.g., the EU AI Act 2024, IPCC AR6 2022) (European Union, 2024; IPCC, 2022) shifted research framing.

## 5.6 The framework as the main contribution

The primary contribution of this paper is the measurement framework, not the specific AI–SDG findings. The framework advances AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed semantic proximity), with explicit marking of Level 3 boundaries (§2.2). The cross-sensitivity transparency (Table 13) is a reusable template: instead of reporting a single gap ranking, it makes the stability of every position transparent across the measurement choices analysts control. The SDG-stratified INLP adaptation — training the corpus classifier on balanced topic–corpus strata so that no linear direction can encode both topic and register — is a methodological contribution applicable to any cross-corpus comparison where topic and register are confounded. The protocol is portable to any research-policy interface that shares a category system and embeds both sides in a common semantic space.

The AI–SDG findings are a first application to be checked against a larger, more diverse policy corpus, not settled results. The findings reported here, the cancellation, the SDG 17 inversion, and the positive adjusted association, are properties of this specific pair of corpora at this moment in time. The framework that produced them is designed to endure.



## 6 Conclusion

The Introduction asked whether shared SDG labels between academic AI research and institutional policy discourse indicate genuine alignment. The answer is no, not without examining framing. This paper demonstrated that coverage divergence (which SDGs each community prioritises) and within-SDG semantic framing divergence (how differently they discuss the same goal) are separable dimensions that can diverge independently: SDG 9 has the second-largest coverage gap but a moderately ranked semantic gap, while SDG 17 has moderate coverage but the widest adjusted gap. The raw correlation between the two dimensions is near-zero (Spearman  $\rho = -0.012$ ), but this masks a register-topic cancellation: after INLP register removal, the adjusted semantic gap is positively associated with the coverage gap ( $\rho = 0.544$ ,  $p = 0.025$ ), confirming that the two communities’ different topical priorities are accompanied by different framings once register is accounted for. The linkage is asymmetric: policy coverage is the strongest directional predictor of framing divergence (positive in 9/9 configurations), while research coverage shows no consistent signal, suggesting that the institutional context in which policy discourse is produced drives the divergence. Where policy attention concentrates, Mode 2 knowledge production (Gibbons et al., 1994) shapes the discourse in contexts of application, producing framings that are structurally further from research norms; this is the expected signature of two distinct knowledge cultures, not a communication failure (Caplan, 1979; Cash et al., 2003).

The practical implication is that topical mention is not alignment. SDG tags, badges, and publication counts capture only whether a text is *about* a goal, not how it frames it. The register decomposition reveals that embedding distance, like keyword matching, is a composite signal: for 16 of 17 SDGs, register uniformly inflates the apparent divergence between research and policy, making the two communities seem more different than their topics warrant. The single exception, SDG 17, is the most instructive case: policy partnership discourse uses institutional language that superficially resembles research vocabulary, masking the deepest topic divergence in the framework. Reporting both dimensions, and acknowledging the register component, would at minimum make the gap visible. The stakes extend beyond academia: funders and policymakers treat SDG tags as evidence of alignment (Heilinger et al., 2024; Heras-Saizarbitoria et al., 2022). This paper shows that assumption is empirically false.

The framework is the paper’s primary contribution. It advances AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed semantic proximity), with explicit marking of Level 3 boundaries (§2.2). The cross-sensitivity grid, the register decomposition, and the coverage–semantic separability test are portable to any research-policy interface that shares a category system and embeds both sides in a common semantic space. Level 3, whether the divergences documented here reflect productive exploration, institutional

failure, or simply different knowledge cultures pursuing different mandates, requires qualitative, institutional, and deliberative methods that lie beyond the scope of embedding analysis. The coverage patterns it captures are externally valid, reproducing the SDG ranking that prior keyword- and citation-based inventories established. The framework does not solve the alignment problem. It makes the problem visible.

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## A Retrieval and Query Chain

The research corpus was retrieved from the OpenAlex API (Priem et al., 2022) using 68 queries formed by the cross-product of OpenAlex’s native SDG work filter and four AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), restricted to `publication_year > 2017`. The 2018 start date reflects the post-AlexNet (Krizhevsky et al., 2012) emergence of deep learning as the dominant paradigm and a research-uptake lag after the SDGs’ adoption in 2015. OpenAlex SDG metadata defines the candidate universe; the SDG assignments used in the analysis are produced by the independently validated supervised classifier (Section 3.5). Complete query construction and credential handling are implemented in `1_code/0_fetch/fetch_openalex.py`.

The retrieval boundary is a deliberate scope condition. Elsevier’s data-driven characterisation of AI research draws on over 700 field-specific keywords across some 600,000 documents (Hellwig et al., 2019), favouring recall. The OECD.AI Observatory methodology instead defines a paper as AI when it carries the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), whose Fields of Study taxonomy OpenAlex later inherited and extended by mapping it to Wikidata identifiers (OECD, 2024; Priem et al., 2022; Wang et al., 2020). The present design prioritises precision: a four-term query keeps non-AI noise out of the embedding space at the cost of recall in specialised AI subfields. Schoeberl et al. (2023) reach the same conclusion for AI-research identification, namely that results are sensitive to the retrieval method and that supervised classification is preferred.

A direct test of this scope condition re-ran retrieval using field-of-study membership for the same two AI fields, mirroring the OECD.AI definition (OECD, 2024), and re-scored the resulting corpus through the identical pipeline. Concept retrieval used OpenAlex concept tags for Artificial Intelligence and Machine Learning, capped at `MAX_PAPERS_CONCEPT_CORPUS = 100,000` papers, yielding the roughly 100,000-paper concept corpus. This scale sits on the empirical variance plateau established by the sample-stability ladder (Appendix F): at 50,000 papers the coverage-profile standard deviation is 0.001 and the mean within-SDG semantic gap is 0.342, within 0.002 of the full-corpus estimate of 0.340 (full corpus  $N = 3,105,144$ ), while policy-text calibration bias has also converged (0.122). The scale also exceeds every corpus size used in prior SDG classification and mapping benchmarks, namely the OSDG community dataset (30,534 texts) Pukelis et al. (2022), the SDG benchmark encoder evaluation corpus (35,001 papers) Gjorgjevikj et al. (2025), the SDGi corpus (20,267 segments) SDGi (2024), the Aurora expert-validated set (4,971 texts) Vanderfeesten et al. (2020), and the 6,000-paper benchmark by Kashnitsky et al. (2024), while remaining far smaller than the full 3,105,144-segment research corpus and thus a computationally tractable substrate for OpenAlex concept tagging.



## B Concept-Retrieval Sensitivity

To test sensitivity to the query choice, retrieval was repeated using field-of-study membership for the two AI fields of study (OECD, 2024), mirroring the OECD.AI Observatory methodology, which treats a paper as AI if tagged with the Artificial Intelligence or Machine Learning field of study in the Microsoft Academic Graph (MAG), the taxonomy that OpenAlex later inherited and extended (Priem et al., 2022; Wang et al., 2020). No SDG filter was applied. SDG assignment was performed by the same trained logistic-regression classifier used throughout (Section 3.5), so the only difference from the main corpus is retrieval strategy. The resulting 100,000-paper corpus was passed through the identical segmentation, embedding, scoring, and analysis pipeline.

Table 4 compares the SDG coverage shares of the keyword-retrieved and concept-retrieved corpora. The two profiles are closely aligned (Spearman  $\rho = 0.948$ , Kendall  $\tau = 0.868$ ); the largest deviations are SDG 4 ( $-7.3\text{pp}$ , the known ML-vocabulary artefact, Appendix C.1) and SDG 9 ( $+5.8\text{pp}$ ). The within-SDG coverage-gap and semantic-gap rankings are also stable against the keyword baseline (coverage-gap rank  $\rho = 0.87$ ; semantic-gap rank  $\rho = 0.81$ , Table 13). Schoeberl et al. (2023) independently find that AI-research identification is sensitive to the retrieval method and recommend supervised classification. The present LR scoring is that recommended step applied to SDG assignment. We therefore report the main findings as robust to the query-term choice, while noting that both corpora remain conditional on their respective retrieval strategies.

**Table 4.** SDG coverage-share comparison: keyword-retrieved versus concept-retrieved (AI/ML field-of-study) research corpus.

| SDG                                              | Keyword % | Concept % | $\Delta$ |
|--------------------------------------------------|-----------|-----------|----------|
| SDG 1                                            | 0.4       | 0.7       | +0.3     |
| SDG 2                                            | 2.4       | 1.8       | -0.6     |
| SDG 3                                            | 25.7      | 25.7      | -0.0     |
| SDG 4                                            | 15.6      | 8.3       | -7.3     |
| SDG 5                                            | 2.0       | 1.7       | -0.2     |
| SDG 6                                            | 1.0       | 0.8       | -0.2     |
| SDG 7                                            | 4.1       | 4.7       | +0.6     |
| SDG 8                                            | 2.2       | 2.3       | +0.1     |
| SDG 9                                            | 20.1      | 25.9      | +5.8     |
| SDG 10                                           | 0.9       | 0.8       | -0.1     |
| SDG 11                                           | 5.1       | 8.0       | +2.9     |
| SDG 12                                           | 1.5       | 2.7       | +1.2     |
| SDG 13                                           | 3.7       | 4.2       | +0.5     |
| SDG 14                                           | 1.6       | 1.3       | -0.3     |
| SDG 15                                           | 2.2       | 2.1       | -0.1     |
| SDG 16                                           | 11.2      | 8.8       | -2.4     |
| SDG 17                                           | 0.3       | 0.2       | -0.0     |
| Spearman $\rho = 0.948$ ; Kendall $\tau = 0.868$ |           |           |          |

*Notes:*  $\Delta$  is concept minus keyword share in percentage points.

## C Diagnostics for Reference Classifier

### C.1 SDG 4 Lexical Artefact Audit: The ML ‘Learning’ Overlap

SDG 4 is the only goal audited because it is the only one carrying an a priori lexical-overlap concern: “learning” is shared ML and education vocabulary, and prior AI-SDG inventories do not rank SDG 4 among the leaders, so its third-place research share is suspect on independent grounds. No other SDG has a comparable overlap. The SDG 4 concern is not used to relabel research records. Its narrower purpose is to test whether the concern is empirically plausible. Table 5 therefore compares the lexical composition of SDG 4-assigned research records with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning vocabulary and education-specific vocabulary.

The audit points to a more nuanced conclusion than a pure false-positive story. More than half of the SDG 4-assigned set contains education-specific vocabulary without ML-only terms, and roughly another third contains both education and ML-learning vocabulary. By contrast, the matched non-SDG4 sample and the SDG 9 comparison set are dominated by ML-only vocabulary. This means the SDG 4 signal is not simply the same technical language that drives SDG 9. However, the large “both” share also confirms that ML-

learning language and education language overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical LR argmax assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap: neither “clean education” nor “pure ML artefact.”

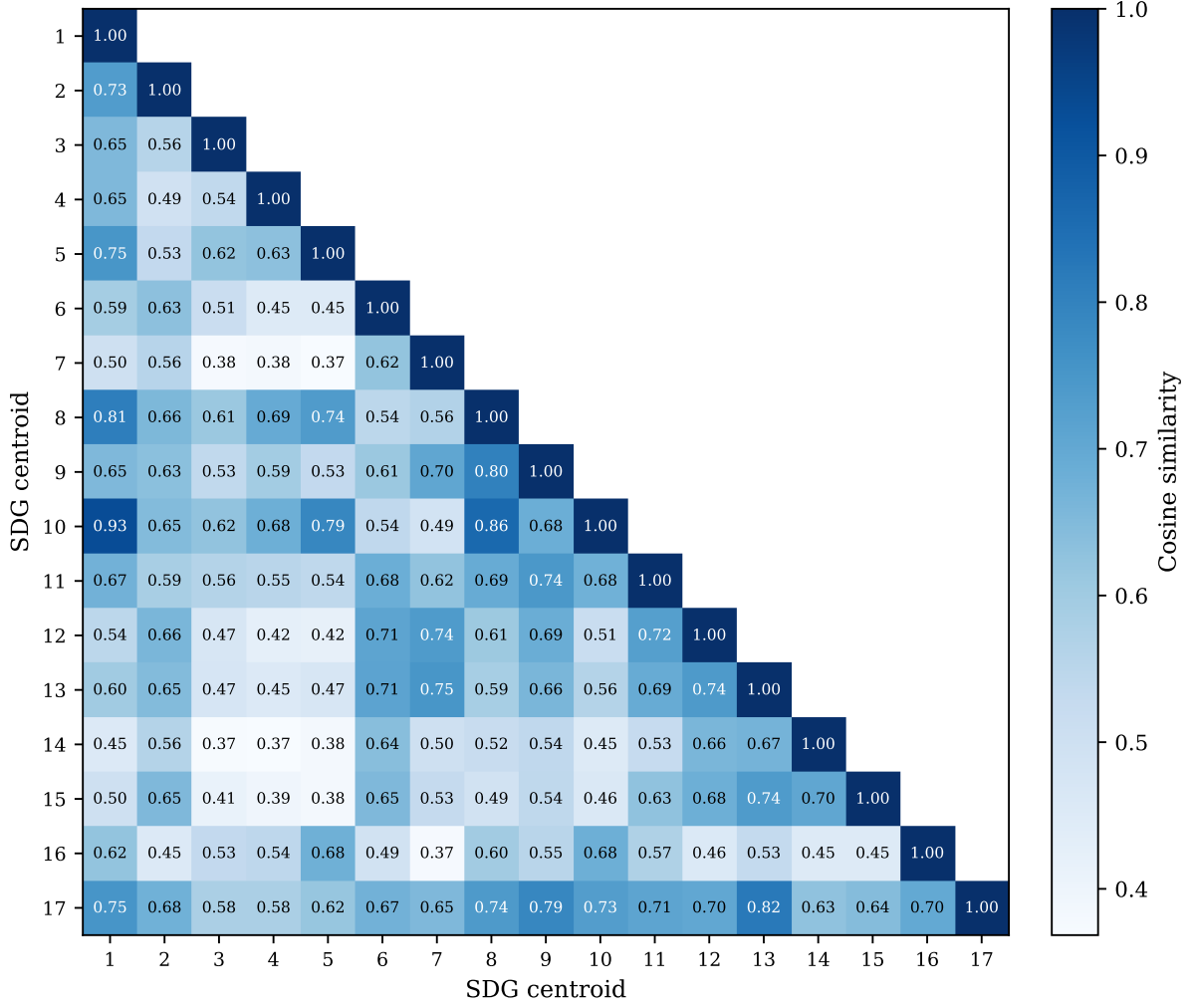
**Table 5.** SDG 4 lexical artefact audit.

| Corpus subset            | N      | ML-only % | Education-only % | Both % | Neither /<br>unclear % |
|--------------------------|--------|-----------|------------------|--------|------------------------|
| SDG 4-assigned research  | 470002 | 11.8      | 44.4             | 23.8   | 19.9                   |
| Non-SDG4 research sample | 470002 | 50.3      | 4.0              | 2.6    | 43.1                   |
| SDG 9-assigned research  | 577833 | 61.5      | 2.5              | 2.9    | 33.1                   |

*Notes:* The audit is not used to reassign SDGs. It only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

## C.2 Centroid Similarity

This subsection reports the pairwise cosine-similarity heatmap (Figure 7) for the 17 canonical SDG centroids. The centroids are the per-SDG mean training embeddings (L2-normalised) used by the LR classifier. The heatmap provides supplementary geometric context for the classification geometry that Section 3.5 builds on.



**Figure 7.** Pairwise cosine similarity between canonical SDG centroids (lower triangle). SDG 13 and SDG 17 sit close together (cosine 0.821), as do SDGs 1, 8, and 10.

Figure 7 shows the full 17×17 similarity grid. The lower triangle is annotated; the upper triangle is left blank because the matrix is symmetric. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the supervised LR classifier (Table 1) can build on this well-separated base to reach 0.816. Second, a small set of off-diagonal cells is elevated. SDG 11’s centroid neighbours SDG 9 (cosine 0.744), and SDG 8 lies near SDGs 1 and 10 (cosine 0.808 and 0.862), and SDGs 1 and 10 are themselves the closest off-diagonal pair (cosine 0.931) – these are the proximities that make the SDG 11 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the clear outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.821), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

## D Lexical Illustration of the Semantic Gap

Table 6 lists the TF-IDF terms most distinctive of the research and policy sides for each SDG. Generic ML methodology vocabulary, such as *model*, *learning*, and *neural*, is excluded because these terms reflect the search strategy rather than SDG-specific framing.

The contrasts echo the embedding patterns. The widest-gap SDGs (3, 13, 11, 7, 1, 8, 14; raw gap  $\geq 0.358$ ) consistently oppose technical measurement vocabulary, *detection*, *images*, *yield*, *regression*, against institutional or welfare terms, *climate*, *emissions*, *health*, *employment*. SDG 13 clarifies the pattern most sharply: research terms centre on crop monitoring and soil science while policy terms address national emissions commitments. Lower-gap SDGs (17, 15, 4, 10, 5, 6, 2; raw gap  $\leq 0.299$ ) show narrower lexical divergence, such as the shared technology-infrastructure frame of SDG 9 or the social-media-partnership overlap of SDG 17. A further observation cuts across both groups: agricultural-AI vocabulary (*crop*, *soil*, *agricultural*) appears on the research side of nine unrelated SDGs, reflecting a strong agri-tech skew in the AI-for-SDG literature rather than alignment with each goal’s policy domain.

**Table 6.** Distinctive TF-IDF terms for all 17 SDGs.

| SDG | Gap   | Research-side distinctive terms                                                                         | Policy-side distinctive terms                                                                                   |
|-----|-------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1   | 0.415 | alleviation, china, approach, zakat, satellite, household, studies, indonesia, literature, relationship | social, cent, national, people, protection, services, population, vulnerable, living, million                   |
| 2   | 0.299 | crop, detection, plant, diseases, images, leaf, precision, techniques, soil, yield                      | population, food, children, prevalence, worst, access, hunger, million, births, ratio                           |
| 3   | 0.494 | patients, cancer, cells, images, clinical, detection, disease, potential, genes, predict                | health, services, care, national, population, births, mortality, rate, people, access                           |
| 4   | 0.267 | cognitive, brain, neurons, sleep, agricultural, food, cells, motor, memory, poor                        | education, school, secondary, primary, national, quality, children, percent, access, vocational                 |
| 5   | 0.278 | menstrual, poor, pain, mothers, differences, relationship, patients, fertility, stress, sex             | gender, violence, equality, national, rights, girls, men, cent, sexual, domestic                                |
| 6   | 0.292 | irrigation, soil, agricultural, crop, salinity, agriculture, sup, potential, sub, yield                 | sanitation, access, drinking, supply, national, wastewater, cent, services, population, clean                   |
| 7   | 0.428 | agricultural, poverty, agriculture, battery, optimization, approach, load, drying, poor, artificial     | access, electricity, renewable, clean, worst best, population, affordable, index worst, cent, national          |
| 8   | 0.387 | detection, image, food, recognition, leaf, plant, cnn, convolutional, disease, techniques               | growth, employment, gdp, worst, labour, best, economic, cent, government, tourism                               |
| 9   | 0.314 | agricultural, agriculture, iot, detection, food, crop, plant, farming, farmers, soil                    | infrastructure, innovation, national, government, public, countries, strategy, access, population, intelligence |
| 10  | 0.271 | wealth, income inequality, effects, evidence, studies, que, distribution, literature, mobility, poor    | people, rights, countries, migration, national, cent, disabilities, persons, discrimination, target             |
| 11  | 0.449 | detection, image, accident, traffic, object, poor, recognition, spatial, fusion, agricultural           | housing, city, urban, transport, cities, public, waste, national, plan, green                                   |
| 12  | 0.296 | food, agricultural, detection, quality, potential, behavior, samples, organic, design, compounds        | waste, consumption, national, recycling, production, procurement, management, public, economy, circular         |
| 13  | 0.480 | soil, agricultural, crop, moisture, drought, sub, rice, stress, agriculture, yield                      | climate, change, emissions, countries, national, action, global, energy, nations, united                        |
| 14  | 0.358 | detection, images, aquaculture, food, bacterial, shrimp, cells, imaging, farming, techniques            | marine, fish caught, biodiversity, rate, ocean, worst, population, sea, embodied, coastal                       |
| 15  | 0.239 | soil, crop, agricultural, plant, spectral, images, sensing, detection, remote, field                    | biodiversity, forest, worst, national, protected, population, area, freshwater, species, best                   |
| 16  | 0.329 | poor, food, covid-, pain, image, detection, social, studies, patients, brain                            | international, nations, peace, rights, united, security, human, law, world, global                              |
| 17  | 0.177 | sentiment, tweets, covid-, twitter, social, agricultural, dan, yang, zakat, media                       | countries, oecd, international, nations, united, agenda, report, financing, global, challenges                  |

Notes: Terms are TF-IDF-weighted unigrams and bigrams from deterministic samples of research and policy texts assigned to each SDG.

## E Distance-Functional Robustness Table

Table 7 reports the per-SDG research–policy gap under the full battery of distribution-aware metrics described in Section E, together with the primary semantic gap. Part 1 covers the full-corpus exact metrics (sliced Wasserstein, Chamfer, Hausdorff, Gaussian-2-Wasserstein, Grassmann, linear MMD); part 2 covers the sampled metrics (exact EMD, Sinkhorn, energy, squared RBF MMD, C2ST AUC) plus the sampled semantic gap. The final row of each part reports Spearman  $\rho$  between each metric’s ranking and the primary (adjusted) ranking.

**Why the raw robustness result does not carry over.** Under raw embeddings the distribution-aware metrics appeared to preserve the semantic gap ranking ( $\rho = 0.44\text{--}0.91$ ). This agreement was driven by the shared *register* component, not by topic. Decomposing the raw gap into adjusted semantic gap + register component, the register component has *larger* variance than the raw gap itself (variance ratio 1.24) and correlates  $\rho = 0.67$  with the raw gap while correlating  $\rho = -0.29$  with the adjusted semantic gap. The distribution-aware metrics therefore tracked register, which is why they agreed with the raw semantic gap but diverge from the adjusted semantic gap. After INLP removes register, the metrics instead track topic-space *shape*, which is why their correlation with the adjusted semantic gap falls to the 0.3–0.7 range and they elevate SDGs 12 and 1 at the margin. The adjusted semantic gap is a mean-direction estimate, and the distribution-aware distances are a complementary full-distribution view. Future work should report both and, where the two disagree, investigate whether the divergence is shape-driven (as here) or a residual artefact.

**Table 7.** Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 1 of 2: full-corpus exact metrics). SWD = sliced Wasserstein;  $\text{MMD}^2$  = squared RBF maximum mean discrepancy; Gauss.  $W_2$  = Gaussian 2-Wasserstein; C2ST = classifier two-sample test AUC. The final row reports Spearman  $\rho$  between each metric’s ranking and the canonical ranking across the 17 SDGs.

| SDG                  | Canonical | SWD   | Chamfer | Hausdorff | Gauss. $W_2$ | Grassmann | linear MMD |
|----------------------|-----------|-------|---------|-----------|--------------|-----------|------------|
| 1                    | 0.191     | 0.009 | 0.509   | 0.600     | 0.684        | 3.727     | 0.067      |
| 2                    | 0.089     | 0.008 | 0.388   | 0.454     | 0.618        | 3.629     | 0.057      |
| 3                    | 0.319     | 0.011 | 0.454   | 0.581     | 0.717        | 3.717     | 0.129      |
| 4                    | 0.158     | 0.010 | 0.399   | 0.487     | 0.607        | 3.307     | 0.096      |
| 5                    | 0.151     | 0.010 | 0.429   | 0.529     | 0.646        | 3.514     | 0.095      |
| 6                    | 0.130     | 0.009 | 0.417   | 0.475     | 0.639        | 3.598     | 0.080      |
| 7                    | 0.170     | 0.010 | 0.427   | 0.522     | 0.687        | 3.729     | 0.105      |
| 8                    | 0.184     | 0.008 | 0.474   | 0.564     | 0.658        | 3.566     | 0.065      |
| 9                    | 0.228     | 0.010 | 0.430   | 0.556     | 0.671        | 3.444     | 0.104      |
| 10                   | 0.195     | 0.008 | 0.456   | 0.517     | 0.617        | 3.343     | 0.062      |
| 11                   | 0.209     | 0.009 | 0.446   | 0.542     | 0.677        | 3.484     | 0.091      |
| 12                   | 0.156     | 0.009 | 0.467   | 0.566     | 0.677        | 3.462     | 0.094      |
| 13                   | 0.219     | 0.010 | 0.417   | 0.498     | 0.667        | 3.667     | 0.117      |
| 14                   | 0.132     | 0.010 | 0.415   | 0.510     | 0.675        | 3.753     | 0.098      |
| 15                   | 0.084     | 0.008 | 0.394   | 0.464     | 0.578        | 3.421     | 0.056      |
| 16                   | 0.242     | 0.010 | 0.417   | 0.520     | 0.640        | 3.333     | 0.103      |
| 17                   | 0.326     | 0.009 | 0.461   | 0.490     | 0.595        | 2.993     | 0.104      |
| $\rho$ vs. canonical | –         | 0.525 | 0.485   | 0.444     | 0.243        | -0.243    | 0.620      |

**Table 8.** Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 2 of 2: sampled metrics; continuation of Table 7). EMD = exact 1-Wasserstein;  $\text{MMD}^2$  = squared RBF-MMD; Centroid (samp.) = sampled centroid-gap control column.

| SDG                  | Canonical | EMD   | Sinkhorn | Energy | $\text{MMD}^2$ | C2ST AUC | Centroid (samp.) |
|----------------------|-----------|-------|----------|--------|----------------|----------|------------------|
| 1                    | 0.191     | 0.645 | 0.623    | 0.056  | 0.025          | 0.969    | 0.186            |
| 2                    | 0.089     | 0.533 | 0.516    | 0.054  | 0.026          | 0.979    | 0.088            |
| 3                    | 0.319     | 0.665 | 0.597    | 0.105  | 0.046          | 0.985    | 0.315            |
| 4                    | 0.158     | 0.554 | 0.523    | 0.084  | 0.040          | 0.988    | 0.157            |
| 5                    | 0.151     | 0.581 | 0.543    | 0.082  | 0.038          | 0.984    | 0.152            |
| 6                    | 0.130     | 0.546 | 0.547    | 0.073  | 0.036          | 0.986    | 0.128            |
| 7                    | 0.170     | 0.581 | 0.545    | 0.094  | 0.045          | 0.990    | 0.170            |
| 8                    | 0.184     | 0.620 | 0.596    | 0.054  | 0.024          | 0.978    | 0.185            |
| 9                    | 0.228     | 0.625 | 0.560    | 0.090  | 0.040          | 0.984    | 0.228            |
| 10                   | 0.195     | 0.602 | 0.636    | 0.051  | 0.023          | 0.964    | 0.191            |
| 11                   | 0.209     | 0.606 | 0.580    | 0.076  | 0.035          | 0.977    | 0.213            |
| 12                   | 0.156     | 0.605 | 0.605    | 0.081  | 0.037          | 0.987    | 0.154            |
| 13                   | 0.219     | 0.577 | 0.485    | 0.102  | 0.048          | 0.983    | 0.217            |
| 14                   | 0.132     | 0.555 | 0.538    | 0.091  | 0.044          | 0.984    | 0.134            |
| 15                   | 0.084     | 0.515 | 0.515    | 0.053  | 0.026          | 0.982    | 0.083            |
| 16                   | 0.242     | 0.604 | 0.495    | 0.088  | 0.039          | 0.983    | 0.241            |
| 17                   | 0.326     | 0.582 | 0.499    | 0.083  | 0.037          | 0.980    | 0.319            |
| $\rho$ vs. canonical | –         | 0.694 | 0.047    | 0.444  | 0.260          | -0.164   | 1.000            |



## F Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, this study reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), 100 independent samples were drawn without replacement, the research coverage profile was recomputed, sampled research centroids were rebuilt, the mean within-SDG semantic gap was recalculated, and the policy-text calibration-bias summary was re-estimated. Policy-side quantities were held fixed. The full 3,105,144-segment analysis is shown as a deterministic anchor row rather than as another repeated sample.

**Table 9.** Sample-stability ladder for the research corpus.

| Sample Size | Mean SD of SDG coverage shares | Mean within-SDG semantic gap | Policy-text calibration bias |
|-------------|--------------------------------|------------------------------|------------------------------|
| 1,000       | 0.006                          | $0.417 \pm 0.020$            | $0.122 \pm 0.008$            |
| 2,000       | 0.004                          | $0.384 \pm 0.015$            | $0.122 \pm 0.006$            |
| 5,000       | 0.003                          | $0.358 \pm 0.009$            | $0.122 \pm 0.004$            |
| 10,000      | 0.002                          | $0.350 \pm 0.006$            | $0.122 \pm 0.002$            |
| 20,000      | 0.001                          | $0.344 \pm 0.005$            | $0.122 \pm 0.002$            |
| 50,000      | 0.001                          | $0.342 \pm 0.003$            | $0.122 \pm 0.001$            |
| 100,000     | 0.001                          | $0.341 \pm 0.003$            | $0.122 \pm 0.001$            |
| 200,000     | 0.000                          | $0.340 \pm 0.002$            | $0.122 \pm 0.001$            |
| 500,000     | 0.000                          | $0.340 \pm 0.001$            | $0.122 \pm 0.000$            |
| 1,000,000   | 0.000                          | $0.340 \pm 0.001$            | $0.122 \pm 0.000$            |
| 2,000,000   | 0.000                          | $0.340 \pm 0.000$            | $0.122 \pm 0.000$            |
| Full corpus | —                              | 0.340                        | 0.122                        |

*Notes:* Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

*Plain-language guide.*<sup>2</sup> The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap ( $\sim 0.340$ ) and policy-text calibration bias ( $\sim 0.122$ ) already stabilise to within 0.001 of their full-corpus values. From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 3,105,144-segment result is retained as the primary estimate because it uses all available data while delivering the tightest values, but the substantive conclusions are not artefacts of sample size choice.

<sup>2</sup>The three metrics are: (1) mean SD of SDG coverage shares, how much the topic balance moves across repeated samples; (2) mean semantic gap, how different research and policy language are within the same SDG; (3) policy-research score gap, how much the scoring instrument favours policy text over research text regardless of content.

The larger corpus also enables per-SDG subgroup detail and tighter rank-order precision that the plateau minimum does not guarantee.

**Balanced-subset rank stability.** The ladder holds the policy side fixed while the research side grows, so within a tier the two sides are never size-matched. Because research (2,536,771 abstracts) outweighs policy (6,367 documents) by about 398:1, this study also tested whether the within-SDG semantic-gap *ranking* survives when the research side is subsampled to a size comparable to the policy corpus. Across 100 independent draws at the 50,000-paper tier, the gap ranking reproduces the full-corpus ordering at Spearman  $\rho = 0.986$  (SD 0.009), already  $\rho = 0.936$  at 10,000 papers and  $\rho = 0.990$  at 100,000 papers, converging to  $\rho = 1.000$  at the full corpus. The research-side over-representation therefore does not determine which SDGs show large versus small semantic gaps.

## G Model Selection: Grid Search Protocol and Rationale

### G.1 Models Tested

Three model families were evaluated under 5-fold GroupKFold cross-validation (document-level grouping, not random splits) on the pooled training data (n=52,835). The test set (n=9,338) was never touched during any grid search.

- **Logistic Regression (LR):** Interpretable per-dimension coefficients per SDG; fast; standard multiclass baseline.
- **MLP (PyTorch):** Can capture non-linear interactions in the embedding space.
- **RF / XGB:** Considered but excluded because the embedding space is already a learned representation – bagged trees add ensemble complexity without a clear mechanism for improvement over LR/MLP, and would not produce interpretable coefficients per dimension.

### G.2 LR Grid Search

The LR grid searched  $C \in \{0.1, 1, 2, 3, 5, 10, 30, 100\}$  with L2 penalty and `class_weight`  $\in \{\text{None}, \text{balanced}\}$  (16 L2 configurations). The best L2 configuration was  $C=3.0$ , `penalty=l2`, `class_weight=None` (CV macro- $F_1 = 0.8101 \pm 0.0020$ ).  $C=1.0$ ,  $C=2.0$  and  $C=5.0$  lie within  $\sim 0.0004$  of the optimum, while  $C=10.0$  is  $\sim 0.0026$  below;  $C=30$  and  $C=100$  degrade clearly (0.804 and 0.797), confirming the optimum is a moderately regularised  $C \approx 3$  and that weaker regularisation overfits.  $C=3.0$  was selected as the empirical optimum. A separate L1 (lasso) spot-check ( $C=10$ ) reached CV macro- $F_1 = 0.8066 \pm 0.0069$ , below the L2 champion, confirming that L1 penalisation is not competitive for this embedding

geometry (prior spot-check documented in 5\_notes/MODEL\_SELECTION.md).

### G.3 MLP Grid Search

The MLP swept  $n\_layers \in \{2, 3, 4, 6\} \times hidden\_size \in \{256, 384\} \times lr \in \{3e-4, 1e-3\}$  (16 configurations) with  $weight\_decay=0$  and  $dropout=0.3$ , trained with AdamW and early stopping. The best MLP (3 layers, 256 hidden,  $lr=3e-4$ ) achieved CV macro- $F_1 = 0.8192 \pm 0.0032$ . Depth 3 was optimal; depth 6 did not improve over depth 3, and depth 2/4 were within  $\sim 0.001$ , so architecture is not the binding constraint. Learning rate had negligible impact: across the two learning rates the CV macro- $F_1$  range at the best architecture was 0.816–0.819.

### G.4 Unified Ranking and Final Selection

| Rank | Config                            | CV macro- $F_1$     |
|------|-----------------------------------|---------------------|
| 1    | MLP 3L/256h (best)                | $0.8192 \pm 0.0032$ |
| 2    | MLP 4L/256h                       | $0.8169 \pm 0.0024$ |
| 3–16 | MLP (remaining 14 configurations) | 0.815–0.819         |
| 17   | LR C=3, L2                        | $0.8101 \pm 0.0020$ |
| 18   | LR C=1, L2                        | $0.8087 \pm 0.0025$ |

**Champion: LR (C=3.0, L2).** The gap between LR and MLP is 0.0091 macro- $F_1$ , a modest difference insufficient to justify the complexity trade-off. MLP requires 3 hidden layers, AdamW, learning rate tuning, early stopping, and dropout selection (16 configurations swept). LR requires one free parameter (C), no hidden layers, no dropout, no random seed sensitivity. For this study’s purpose – building an interpretable supervised reference to compare corpora, not a classifier built to label single documents – LR is the right choice. LR provides 768 signed coefficients per SDG class, each directly attributable to a specific semantic dimension; MLP’s hidden layers diffuse this information across non-linear transformations. The macro- $F_1$  numbers are reported honestly: LR  $0.8101 \pm 0.0020$ , MLP  $0.8192 \pm 0.0032$ , gap 0.0091 macro- $F_1$ . LR is chosen for transparency, not superior accuracy.

### G.5 Held-Out Test Evaluation of the MLP Robustness Check

The MLP test macro- $F_1$  (0.826) cited in Section 4.1 is a separate evaluation from the CV grid-search results above. After the CV sweep, the champion MLP architecture (3 layers, 256 hidden,  $lr=3e-4$ ) was retrained on the full training pool ( $n=52,835$ ) and evaluated once against the held-out test set ( $n=9,338$ ), the same sanctioned single-use test evaluation applied to the LR champion. The resulting test macro- $F_1$  of 0.826 is not

directly comparable to the CV mean of  $0.8192 \pm 0.0032$  from the grid search: the CV number is an average across five document-grouped folds on the training pool alone, while the test number is a single evaluation on a disjoint held-out sample after retraining on the full pool. Both numbers confirm that the LR–MLP gap is modest and that architecture choice is not the binding constraint on classification quality.

# H Register Removal: Convergence and Validation

## H.1 Convergence

The full INLP procedure and identification argument are given in Methodology (Section 3.8). This appendix reports the convergence behaviour of the procedure and the construct validation of the component it removes. The adjusted results are the primary decomposition, not sensitivity diagnostics.

**Convergence results.** Held-out accuracy dropped from 0.978 (iteration 1) to 0.498 by iteration 62, at which point no further linear register separation was detectable. The Spearman rank correlation between the iteration 1 and iteration 62 gap vectors was 0.343: the additional removed directions shifted the relative SDG ordering substantially. The mean gap compresses from 0.340 to 0.187, a reduction of 40%, before plateauing. The Spearman  $\rho$  decays rapidly in the first 15 iterations (1.000 to 0.456) and then stabilises around 0.369, indicating that the first 15 orthogonal directions capture the bulk of the register signal.

The decomposition reveals a striking pattern for SDG 17 (Partnerships for the Goals). Under the raw semantic gap, SDG 17 shows the smallest gap (0.177, rank 17 of 17), suggesting near-identical framing. After register removal, the adjusted gap is 0.326, the largest adjusted gap in the corpus (rank 1). The register component is -0.149, the only negative value, indicating that register actually *increases* the semantic similarity for this policy-dominant SDG: the policy discourse on partnerships uses register conventions that make it appear closer to research than the underlying topic content warrants. For all other SDGs, the register component is positive (range: +0.155 to +0.261), confirming that register uniformly inflates the raw gap. The headline correlation pattern is consistent with the apparent cancellation: the raw coverage-vs-gap correlation is near-zero ( $\rho = -0.012$ ,  $p = 0.959$ ) because the register component ( $\rho = -0.390$ ,  $p = 0.120$ ) appears to cancel the topic signal ( $\rho = 0.544$ ,  $p = 0.025$ ; Table 2).

**Cross-config replication.** The convergence pattern is not specific to the canonical MPNet track. The same iterative procedure was run on the MiniLM and SciBERT encoder tracks (the latter two on the S1 subset, by design); all three reach the  $\leq 0.5$  stopping threshold. Held-out register-classification accuracy falls from 0.965–0.978 at iteration 1 to 0.491–0.498 at convergence (MiniLM: 40 iterations, final 0.498; SciBERT: 79 iterations, final 0.491; MPNet: 62 iterations, final 0.498), and all three compress sharply in the first 15 iterations (to 0.539–0.685) before plateauing at the majority-class baseline. The iteration count required to reach the threshold differs across encoders, but the qualitative convergence, a rapid early drop followed by a plateau, is identical, confirming that the

stopping behaviour reported above is a property of the INLP procedure rather than of the MPNet embedding space.

## H.2 Iteration-Level Diagnostic

Table 10 shows the per-iteration diagnostic: held-out classification accuracy, mean within-SDG gap, and rank correlation with the iteration 1 gap vector. The accuracy drops sharply in the first 15 iterations and then plateaus, confirming that the first 15 orthogonal directions capture the bulk of the register signal. The mean gap compresses and stabilises at a non-zero value, demonstrating that the semantic gap is not an artefact of corpus-level register differences.

**Table 10.** Stratified iterative register-removal diagnostic.

| Iteration | Test acc. | Mean gap | Spearman $\rho$ vs iter 1 |
|-----------|-----------|----------|---------------------------|
| 1         | 0.978     | 0.315    | 1.000                     |
| 2         | 0.938     | 0.304    | 0.988                     |
| 3         | 0.895     | 0.290    | 0.985                     |
| 4         | 0.864     | 0.282    | 0.983                     |
| 5         | 0.831     | 0.279    | 0.976                     |
| 10        | 0.747     | 0.238    | 0.721                     |
| 15        | 0.685     | 0.206    | 0.456                     |
| 20        | 0.593     | 0.195    | 0.397                     |
| 30        | 0.558     | 0.188    | 0.400                     |
| 40        | 0.535     | 0.186    | 0.360                     |
| 50        | 0.523     | 0.187    | 0.343                     |
| 62        | 0.498     | 0.187    | 0.343                     |

*Notes:* Test acc. = held-out accuracy on a stratified 15% test split after projecting out the first  $k$  learned directions. Mean gap = average within-SDG semantic gap after  $k$  removals. Spearman  $\rho$  = rank correlation of the gap vector at iteration  $k$  versus iteration 1.

### H.3 Construct Validation: Purpose and Validation Question

The register adjustment of Section 3.8 removes, through Iterative Nullspace Projection, the subspace of the embedding space that is linearly decodable as corpus identity. The structural argument for reading that subspace as register is that every INLP iteration is trained on SDG-balanced research and policy embeddings: no linear direction can simultaneously encode SDG topic and corpus identity, so what remains linearly decodable is the corpus-level register difference. That argument guarantees *what* is removed, a corpus-separable linear component, but it does not by itself establish what that component is. This appendix evaluates the register interpretation empirically, using an independent surface-linguistic register score computed from the segment texts, following Biber-style register features (Biber, 1988).

The validation can establish three things: whether the adjustment removes a large corpus-separable component; whether the removal is accompanied by evidence that SDG (topic) structure is being destroyed; and whether apparent residual register signals survive controlled samples. It cannot establish that the removed subspace is identical to the measured linguistic register features. The register score used here is a small set of surface indicators, and the validation is single-encoder and sample-limited. The conclusion is correspondingly bounded: the evidence is consistent with a register reading and provides partial, not complete, empirical support for it.

### H.4 Construct Validation: Register Operationalisation and Sample Construction

**Sample construction.** All results use the canonical MPNet track, 408 segments per sample (12 per SDG per corpus, seed 42), and the same frozen projection matrix  $G$  as the main analysis. A small number of long flagship reports (the SDSN Sustainable Development Reports 2024 and 2025, the UNDP Human Development Report 2021/22, the WHO report on ethics and governance of AI for health, the EU AI Act, and two UN SDG progress reports) each contribute many near-duplicate segments: 7 source documents, global flagship reports (the SDSN Sustainable Development Reports 2024 and 2025, the UNDP Human Development Report 2021/22, the WHO report on ethics and governance of AI for health, the EU AI Act, and two UN SDG progress reports), contribute 25 segments across 15 of the 17 SDGs. These segments are register outliers (mean sentence length 276.9 words vs. 35.6 for the remaining units; passive rate 3.3 vs. 9.8; first-person rate 19.0 vs. 5.9; nominalisation rate 20.0 vs. 38.8), and because segments of the same document sit close together in embedding space they can create spurious pooled correlations.

The primary validation therefore uses a one-per-parent design (global deduplication by source document; 408 distinct parents, no multi-parent units), which caps each flagship

document at a single segment and replaces excluded flagship segments with other policy segments from the same SDG, drawn predominantly from shorter national and thematic monitoring reports. The two designs differ partly in composition, which the Limitations return to.

**Register score.** For every sampled segment, six register features are computed per 1,000 words: hedge-word rate, deontic-modality rate, passive-voice frequency (a be-form immediately preceding a past participle, allowing one intervening adverb), mean sentence length, first-person-pronoun rate, and nominalisation rate; and the first principal component of the standardised features is taken as the register score. The orientation (positive loading on sentence length) is a sign convention only; the score’s interpretation does not rest on any single feature, and sentence length in particular is composition-sensitive, as shown below.

**Corpus feature contrasts.** With the sample structure established, the feature differences can be read correctly. Policy segments are markedly more deontic (3.130 vs. 0.780 per 1,000 words) and research segments hedge more (1.427 vs. 0.854) and passivise more (10.303 vs. 8.453); these contrasts are stable across samples and point in the directions commonly associated with institutional-modality and academic writing register, respectively. First-person and nominalisation rates are roughly balanced at the corpus level (6.463 vs. 6.901; 39.003 vs. 36.367), the flagship segments being the outliers on both. Mean sentence length is the one feature whose corpus direction is not robust: policy segments appear much longer on the original sample (63.803 vs. 37.035 words), but this contrast is driven by the flagship-document segments, whose extraction-artefact-inflated mean (276.9 words) does not generalise, and it reverses direction once those segments are set aside. Sentence length was therefore treated as composition-sensitive and used to motivate the robustness checks below rather than as a load-bearing indicator of register.

## H.5 Construct Validation: Corpus Discrimination Before and After INLP

A 5-fold logistic-regression classifier distinguishes research from policy segments almost perfectly on raw embeddings (accuracy 0.909 on the original sample; 0.944 on the one-per-parent sample; Table 11). After register removal the accuracy falls to 0.505 on the original sample (95% CI [0.457, 0.553], binomial  $p = 0.441$  against 0.5) and to 0.603 on the one-per-parent sample (95% CI [0.555, 0.649],  $p = 1.9e - 05$ ). The adjustment therefore removes a substantial share of the corpus-linear signal. The removal is not complete: the primary one-per-parent value is significantly above chance.

A per-SDG-deduplicated variant of the sample (dominated by the 25 mega-document policy segments above) yields an adjusted accuracy of 0.505, indistinguishable from chance;



excluding those segments raises it to 0.574 (difference +0.098, 95% CI [0.025, 0.169]), confirming the mega-document clustering drives the original-sample result. The canonical one-per-parent estimate (0.603) is the primary result; residual above-chance separation after removal is not a method failure, since INLP’s stopping rule converges on its own training distribution while the validation classifier is re-estimated from scratch, so the two evaluations are not directly comparable.

The register-only rows of Table 11 bound the interpretation. A classifier trained on the six surface features alone reaches only 0.456 on the original sample and 0.544 on the one-per-parent sample, barely above chance in the latter case. The linguistic markers therefore capture only a part of the corpus distinction. This is consistent with the register reading (the corpora do differ in register-like surface ways), but it also delimits it: the removed embedding component is far broader than these six surface indicators, so the validation cannot claim that the removed subspace is exhausted by, or identical to, the measured features.

## H.6 Construct Validation: Topic Preservation

If the adjustment removed topic, 17-way SDG selectivity, how well the SDG of a segment can be predicted from its embedding, would fall. It does not (Table 12): on the original per-SDG-dedup sample (draw 1), LR cross-validated accuracy is 0.691 on raw and 0.672 on adjusted embeddings, and kNN 0.554 and 0.578 (chance 0.059). The change is small and of mixed sign, and is best read as sampling noise around an unchanged quantity rather than as evidence of preservation in the strong sense. The correct statement is that there is no evidence that topic structure is systematically removed; the analysis does not claim that topic is preserved perfectly, and it was run on the original sample only, not re-estimated on the one-per-parent rebuild. Topic remains linearly decodable after register removal, as the identification argument anticipates.

## H.7 Construct Validation: Residual Register Diagnostics and Robustness Investigation

Two checks examine whether residual register survives adjustment. First, the register score correlates positively with the amount of embedding removed on the per-SDG-deduplicated sample (+0.102,  $p = 0.040$ ; within-research  $\rho = +0.212$ , within-policy  $\rho = +0.191$ ). Second, distance to the SDG centroid in adjusted space correlates more strongly with the register score than in raw space (raw  $\rho = +0.126$  rising to adjusted  $\rho = +0.247$ ). Both signals are absent on the controlled one-per-parent sample: the removed-norm correlation becomes  $\rho = +0.092$  (n.s.; within-corpus -0.043 and -0.036, both n.s.), and the centroid-distance correlation becomes negative (raw  $\rho = -0.212$ ; adjusted  $\rho = -0.197$ ).

The original patterns are attributable to flagship-document clustering, register-extreme, mutually similar segments of a few documents dominating the pooled correlation, not to residual register, so the direct surface-linguistic corroboration of the removed subspace is partial: the register reading rests on the corpus-discrimination result and the absence of counter-evidence rather than a measured correspondence between the removed component and the six features.

A within-SDG decomposition (register score vs. distance to the *other-corpus* SDG centroid, adjusted space) yields one nominally surviving trace among policy segments on the initial draw ( $\rho = -0.197$ ,  $p = 0.005$ ; this value reproduced exactly on a re-run of the same sample and survived mega-document exclusion on that draw,  $\rho = -0.213$ ,  $p = 0.003$ ). A draw-stability check re-samples the one-per-parent construction with fresh random streams (seeds 43–45): the statistic is draw-unstable:  $\rho = -0.130$  ( $p = 0.063$ ),  $\rho = -0.004$  ( $p = 0.949$ ),  $\rho = +0.126$  ( $p = 0.070$ ), with the sign flipping across draws and the mean near zero; and mega-document exclusion is likewise draw-specific ( $\rho = +0.127$  n.s.,  $\rho = -0.138$  n.s.,  $\rho = -0.213$   $p = 0.003$  on draws 1–3). Per-SDG, only 0 of 17 reach  $p < 0.05$  (14 negative, 3 positive). The trace is therefore draw-level noise, not residual register structure.

## H.8 Construct Validation: Validation Conclusion

The validation supports the following claims. (i) The adjustment removes a substantial corpus-level component: corpus accuracy falls from 0.909 to 0.505 on the original sample and from 0.944 to 0.603 on the one-per-parent sample. (ii) The removed component is not strongly attributable to SDG topic loss: 17-way SDG selectivity is essentially unchanged. (iii) Apparent residual register signals were explained by sampling and document-clustering effects rather than by stable residual structure: the initial removed-magnitude and centroid-distance associations vanished or reversed on the controlled sample, and the one surviving within-SDG trace was draw-unstable. (iv) The remaining evidence provides partial empirical support for interpreting the removed component as register-related: the corpora differ in register-like surface features in stable directions, and the adjustment removes corpus-separable signal while leaving topic structure intact, but the direct correspondence between the removed component and the measured linguistic markers was not established, and residual corpus-linear signal remains after removal.

The register interpretation is therefore consistent with the validation evidence rather than confirmed by it. The conclusion should be read alongside the main text, which describes the register interpretation as resting on partial, not complete, empirical support.

## H.9 Construct Validation: Limitations

The validation is bounded in ways that should be read alongside the main text. First, the register score is operationalised as the first principal component of six hand-specified features, and the choice of operationalisation is not resolved here: an alternative a-priori “institutional” z-score combination gave null or reversed results on the initial screen, and the score is oriented by a sign convention rather than by a substantive prior. Second, each sample contains only 408 segments (12 per SDG per corpus), and the per-SDG tests within the draw-stability stage have  $n = 12$  policy units each, so the 0-of-17 significant per-SDG result is not evidence of absence at the per-SDG level. Third, the draw-stability check was applied to the within-SDG trace only; the corpus-level accuracies in Table 11 were not re-estimated on fresh draws. Fourth, the one-per-parent sample is a rebuild, not a subset of the original sample, and the excluded flagship documents are replaced by shorter national and thematic monitoring documents, so the two-sample comparison is partly compositional; the mega-vs-non-mega feature contrast documents that shift, and the selectivity check was run on the original sample only. Fifth, the validation is MPNet-canonical only. Sixth, the residual corpus-linear signal that survives adjustment is not characterised further here; its exact linguistic content remains open, which is part of why the empirical support is partial rather than complete. Finally, two distinct statistics in this appendix round to  $-0.197$  at three decimals: the one-per-parent pooled centroid-distance correlation ( $-0.197$ ) and the draw-stability-stage policy other-corpus distance correlation ( $-0.197$ ); they are different quantities and should not be conflated.

**Table 11.** Corpus discrimination accuracy (research vs. policy, 5-fold logistic regression) before and after INLP register removal.

| Sample                   | Input                          | $n$ | Fold-mean acc. | Pooled acc. ( $k/n$ ) | Wilson 95% CI  | Binom. $p$ |
|--------------------------|--------------------------------|-----|----------------|-----------------------|----------------|------------|
| Original (per-SDG dedup) | register-only                  | 408 | 0.456          | 0.456 (186/408)       | [0.408, 0.504] | 0.967      |
| Original (per-SDG dedup) | raw embeddings                 | 408 | 0.909          | 0.909 (371/408)       | [0.878, 0.933] | < 0.001    |
| Original (per-SDG dedup) | adjusted (INLP)                | 408 | 0.505          | 0.505 (206/408)       | [0.457, 0.553] | 0.441      |
| One-per-parent           | register-only                  | 408 | 0.544          | 0.544 (222/408)       | [0.496, 0.592] | 0.042      |
| One-per-parent           | raw embeddings                 | 408 | 0.944          | 0.944 (385/408)       | [0.917, 0.962] | < 0.001    |
| One-per-parent           | adjusted (INLP)                | 408 | 0.603          | 0.603 (246/408)       | [0.555, 0.649] | < 0.001    |
| Original (per-SDG dedup) | adjusted, mega-policy excluded | 383 | 0.574          | 0.574 (220/383)       | [0.524, 0.623] | 0.002      |

*Notes:* Pooled accuracy = proportion of correct predictions across the five test folds. Wilson 95% CI and one-sided binomial test against 0.5 on the pooled proportion. In the one-per-parent sample each segment comes from a distinct source document, so no test segment shares a source document with its training folds; in the original per-SDG-dedup sample, segments of the same flagship document can straddle folds, so the original-sample rows carry a mild grouped-similarity caveat. Mega-policy excluded = original sample minus the 25 policy segments contributed by the 7 multi-segment documents.

**Table 12.** 17-way SDG (topic) selectivity before and after register removal (original per-SDG-dedup sample, draw 1).

| Input           | LR CV acc. | kNN(5) CV acc. | Chance |
|-----------------|------------|----------------|--------|
| Raw             | 0.691      | 0.554          | 0.059  |
| Adjusted (INLP) | 0.672      | 0.578          | 0.059  |

*Notes:* Five-fold stratified cross-validated accuracy of logistic regression ( $C = 10$ ) and  $k$ -nearest-neighbour classification ( $k = 5$ ) (Cover & Hart, 1967) in predicting the SDG of each segment from raw and adjusted embeddings; chance =  $1/17$ . If the adjustment removed topic, these accuracies would fall.

## I Supplementary Cross-Method Data

### I.1 Cross-Method Coverage and Semantic Gap Values

Tables 19 and 20 report the raw coverage gap and semantic gap values for each SDG across seven encoder–assignment–method combinations, with an additional concept-retrieval LR column for comparison against the keyword-retrieved research corpus baseline (Appendix B). While Tables 13 and 18 rank these values to compare cross-method stability, the tables below show the unrounded gap magnitudes, allowing the rank separation to be interpreted in absolute terms. Coverage gap is document-weighted (Assumption A19); semantic gap policy centroids use a 50-segment-per-document cap. The zero-shot nearest-centroid method is reported for the canonical MPNet encoder only; it is scoped to a single supervised-versus-nearest-centroid comparison (Appendix I.4), not a cross-encoder axis.

### I.2 Rank Robustness: Cross-Sensitivity and Encoder Sensitivity

Tables 13 and 14 report each SDG’s semantic-gap rank under the canonical MPNet–LR configuration and six alternatives spanning policy source family (curated AI/SDG, SDGi, UNGDC), segment cap (20, none), and retrieval strategy (concept-based field-of-study). Tables 15, 16, and 17 report the within-SDG gap rankings under MPNet, MiniLM, and SciBERT for each assignment method. The adjusted (register-removed) ranking is reported separately from the raw (naive) ranking, with the tested dimension nested inside each table.

### I.3 Encoder Sensitivity

The semantic-gap ranking is moderately preserved across the three encoders (MPNet–LR vs. SciBERT–LR Spearman  $\rho = 0.54$ ; MPNet–LR vs. MiniLM–LR  $\rho \approx 0.63$ ), and the coverage-gap ranking is highly stable (MPNet–LR vs. SciBERT–LR  $\rho = 0.93$ ). MiniLM and SciBERT are scored on a deterministic 100,000-paper subset (seed 42) of the canonical segmented corpus, so the architecture comparison isolates encoder choice from corpus scale.

Under the raw gap, SDG 3 leads for MPNet-LR while SDG 13 leads for MiniLM-LR and SciBERT-LR; SDG 17 trails under all three. The scientific-domain SciBERT compresses the magnitudes but does not change which goals diverge most. Assignment-method sensitivity within MPNet: the three assignment methods, LR, zero-shot nearest-centroid, and the MLP robustness check, show varied agreement (MPNet LR vs. MLP  $\rho = 0.85$ , LR vs. zero-shot  $\rho = 0.60$ ). SDG 17 is the most method-sensitive goal, sitting at rank 17 under LR but rank 8 under zero-shot. The LR baseline is the canonical reference, with MLP and zero-shot as uncertainty bounds.

## I.4 Assignment-Method Comparison: Supervised vs. Nearest-Centroid

The zero-shot nearest-centroid method is used in this study for one purpose only: a single comparison between the canonical supervised assignment (LR, with the MLP as robustness) and an assignment rule with no trained decision boundary. Under the zero-shot rule, every research paper and policy segment is assigned to the SDG whose train-split reference centroid is most similar (cosine) in embedding space. Table 21 reports how often the two methods agree, per SDG and overall, for the research corpus (paper level) and the policy corpus (segment level), together with the LR-versus-zero-shot semantic-gap rank delta per SDG.

The comparison confirms that the two methods agree on the majority of assignments: the overall LR-versus-zero-shot agreement rate is 67.3% over the 3,105,144 research segments and 81.5% over policy segments (81.4% at document level; Table 21). Agreement is not uniform across SDGs, and the disagreements are concentrated where the reference centroids are least separated, in line with the centroid-similarity diagnostics (Appendix C.2). SDG 17, whose reference centroid is the flattest and most isolated in that analysis, is also the most method-sensitive goal here: it shows the largest rank delta between the two semantic-gap rankings ( $|\Delta| = 9$ ) and one of the lowest per-SDG assignment agreement rates on the research side (26.1%, second only to SDG 10 at 23.6%). SDG 8, whose supervised rank is also heavily method-dependent (rank 6 under LR, rank 14 under zero-shot), shows one of the largest rank deltas ( $|\Delta| = 8$ , tied with SDG 9). The zero-shot rule, having no supervised decision boundary, anchors partnership and inequality discourse less sharply than the LR classifier. The MLP robustness check agrees with the zero-shot rule at 76.3% of policy segments (79.9% at document level), close to the LR-versus-zero-shot rate, and at 59.5% of research segments overall.

## I.5 Raw-Value Correlation: Metric-Sensitivity Companion

Table 22 repeats the H1a–H1d grid of Table 3 with one change: cells report Pearson  $r$  on the raw coverage and gap magnitudes instead of Spearman  $\rho$  on their ranks; the grid, predictors, and columns are identical. The four H1 hypotheses are co-equal and each probes a distinct coverage-structure prediction. Under both metrics the adjusted-gap association is stable across the supervised tracks for H1a and H1d (the hypotheses with a detectable signal: positive in 7/9 and 9/9 configs respectively under Pearson) and remains null for H1b and H1c (no consistent signal under either metric – itself informative: coverage divergence tracks the gap and policy share, not dominance or raw research share). The only reversals are the two SciBERT configs; the zero-shot config, already scoped as an uncertainty bound, is stable on the adjusted gap. SciBERT is designated a lower-trust robustness encoder (its scientific pretraining compresses gap magnitudes, 0.04 against 0.34, Section I.3).

*Possible causes of the SciBERT divergence.* (A) At  $n = 17$  the Pearson estimates are noisy and the permutation null is only rejected for large magnitudes: both SciBERT adjusted-gap values are small in absolute terms and do not reach significance, so the rank/magnitude sign flip is within estimation error. (B) Pearson weights absolute distances, Spearman weights ranks; SciBERT’s  $\sim 8\times$  smaller gap magnitudes make Pearson disproportionately sensitive to its few larger gaps. (C) SciBERT has the lowest assignment fidelity of the three encoders, giving noisier per-SDG gaps. (D) The flip is in the *adjusted* gap, and INLP removal is already documented as encoder-dependent. (E) Pearson is leveraged by individual outliers (e.g. SDG 3). None of A–E is established; A and E are the most directly testable. All are consistent with the divergence being uninformative for the conclusion, which rests on the supervised tracks where the signal is large (+0.70) and stable across metrics. The raw-value specification therefore confirms, rather than threatens, the rank-based result.

## I.6 Detailed Policy Profiles by Source Family

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Table 23 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the SDGi and UNGDC families are more strongly concentrated on climate, partnerships, and governance, consistent with the UNGD

topic distributions reported by Jankin et al. (2024), and, in the SDGi case, SDG 11-style institutional review language. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and SDG 17 remains the largest semantic-gap case across all three families.

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

The symmetric construction preserves the cancellation at the level of the interaction test. The H1a–H1d tests ask whether SDG coverage measures predict the within-SDG semantic gap (Section 4.3). Re-running those tests within each policy source family (Table 24) reproduces the full-corpus null under the curated construction: with policy restricted to the curated AI/SDG family, research share does not correlate with the gap ( $\rho = -0.06$ ,  $p = 0.833$ ), and the coverage-gap correlation also does not reach significance ( $\rho = -0.22$ ,  $p = 0.386$ ). The full-corpus row of Table 24 reproduces the canonical interaction estimates (Section 4.3), confirming the replication is faithful. The null is therefore not an artefact of comparing an AI-specific research intersection with a broader policy union.

**Table 13.** Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

| SDG                  | Canon | Segment cap |      | Retrieval |      |
|----------------------|-------|-------------|------|-----------|------|
|                      |       | 20          | None | LR        | MLP  |
| SDG 1                | 8     | 8           | 8    | 4         | 6    |
| SDG 2                | 16    | 16          | 17   | 17        | 17   |
| SDG 3                | 2     | 1           | 2    | 3         | 5    |
| SDG 4                | 11    | 11          | 11   | 14        | 14   |
| SDG 5                | 13    | 13          | 13   | 6         | 10   |
| SDG 6                | 15    | 15          | 15   | 15        | 16   |
| SDG 7                | 10    | 10          | 9    | 13        | 13   |
| SDG 8                | 9     | 9           | 7    | 5         | 1    |
| SDG 9                | 4     | 5           | 3    | 7         | 7    |
| SDG 10               | 7     | 7           | 10   | 9         | 3    |
| SDG 11               | 6     | 6           | 5    | 2         | 2    |
| SDG 12               | 12    | 12          | 12   | 10        | 11   |
| SDG 13               | 5     | 4           | 4    | 11        | 12   |
| SDG 14               | 14    | 14          | 14   | 12        | 9    |
| SDG 15               | 17    | 17          | 16   | 16        | 15   |
| SDG 16               | 3     | 3           | 6    | 8         | 8    |
| SDG 17               | 1     | 2           | 1    | 1         | 4    |
| Rank Corr ( $\rho$ ) | 1.00  | 1.00        | 0.97 | 0.75      | 0.68 |

*Adjusted (register-removed, canonical).*

*Notes:* Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap). Panel (a) reports the register-removed (adjusted) ranking, which is the canonical measure of this study; Panel (b) reports the naive baseline (un-adjusted, raw) ranking. Within each panel, the Canon column is the canonical MPNet—LR ranking. Policy-source columns compare the keyword-retrieved research profile against each policy-source family. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr ( $\rho$ ) is the Spearman correlation of each column’s SDG gap ranks against that panel’s Canon column.



**Table 14.** Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

| SDG                  | Policy source |         |      |       |      | Segment cap |      | Retrieval |      |
|----------------------|---------------|---------|------|-------|------|-------------|------|-----------|------|
|                      | Canon         | Curated | SDGi | UNGDC | Full | 20          | None | LR        | MLP  |
| SDG 1                | 5             | 2       | 5    | 5     | 5    | 5           | 5    | 1         | 1    |
| SDG 2                | 10            | 9       | 11   | 11    | 10   | 10          | 9    | 11        | 10   |
| SDG 3                | 1             | 8       | 1    | 1     | 1    | 1           | 2    | 3         | 7    |
| SDG 4                | 15            | 15      | 12   | 15    | 14   | 15          | 14   | 13        | 17   |
| SDG 5                | 13            | 13      | 14   | 13    | 13   | 13          | 13   | 6         | 5    |
| SDG 6                | 12            | 10      | 13   | 12    | 12   | 12          | 12   | 15        | 15   |
| SDG 7                | 4             | 1       | 6    | 2     | 4    | 4           | 4    | 8         | 11   |
| SDG 8                | 6             | 5       | 7    | 6     | 6    | 6           | 6    | 4         | 2    |
| SDG 9                | 9             | 14      | 2    | 10    | 9    | 8           | 8    | 12        | 13   |
| SDG 10               | 14            | 11      | 15   | 16    | 15   | 14          | 15   | 10        | 8    |
| SDG 11               | 3             | 4       | 3    | 4     | 3    | 3           | 3    | 2         | 3    |
| SDG 12               | 11            | 6       | 10   | 8     | 11   | 11          | 11   | 14        | 12   |
| SDG 13               | 2             | 3       | 4    | 3     | 2    | 2           | 1    | 5         | 4    |
| SDG 14               | 7             | 7       | 9    | 9     | 7    | 7           | 7    | 9         | 6    |
| SDG 15               | 16            | 12      | 16   | 14    | 16   | 16          | 16   | 16        | 16   |
| SDG 16               | 8             | 16      | 8    | 7     | 8    | 9           | 10   | 7         | 9    |
| SDG 17               | 17            | 17      | 17   | 17    | 17   | 17          | 17   | 17        | 14   |
| Rank Corr ( $\rho$ ) | 1.00          | 0.74    | 0.91 | 0.96  | 1.00 | 1.00        | 0.99 | 0.81      | 0.68 |

*Raw (naive baseline, un-adjusted).*

*Notes:* Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap). Panel (a) reports the register-removed (adjusted) ranking, which is the canonical measure of this study; Panel (b) reports the naive baseline (un-adjusted, raw) ranking. Within each panel, the Canon column is the canonical MPNet—LR ranking. Policy-source columns compare the keyword-retrieved research profile against each policy-source family. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr ( $\rho$ ) is the Spearman correlation of each column’s SDG gap ranks against that panel’s Canon column.

**Table 15.** Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

| SDG                  | Encoder (embedding architecture) |           |           |                |           |                 |           |
|----------------------|----------------------------------|-----------|-----------|----------------|-----------|-----------------|-----------|
|                      | MPNet (768-d)                    |           |           | MiniLM (384-d) |           | SciBERT (768-d) |           |
|                      | LR                               | MLP       | ZS        | LR             | MLP       | LR              | MLP       |
| SDG 1                | <b>8</b>                         | <b>11</b> | <b>1</b>  | <b>8</b>       | <b>7</b>  | <b>15</b>       | <b>1</b>  |
| SDG 2                | <b>16</b>                        | <b>16</b> | <b>14</b> | <b>17</b>      | <b>17</b> | <b>17</b>       | <b>16</b> |
| SDG 3                | <b>2</b>                         | <b>1</b>  | <b>3</b>  | <b>2</b>       | <b>1</b>  | <b>5</b>        | <b>12</b> |
| SDG 4                | <i>11</i>                        | <i>14</i> | <i>13</i> | <i>7</i>       | <i>8</i>  | <i>9</i>        | <i>11</i> |
| SDG 5                | <i>13</i>                        | <i>10</i> | <i>11</i> | <i>1</i>       | <i>3</i>  | <i>2</i>        | <i>6</i>  |
| SDG 6                | <b>15</b>                        | <b>15</b> | <b>10</b> | <b>16</b>      | <b>15</b> | <b>14</b>       | <b>17</b> |
| SDG 7                | 10                               | 12        | 8         | 12             | 12        | 7               | 10        |
| SDG 8                | 9                                | 4         | 12        | 11             | 11        | 13              | 14        |
| SDG 9                | <i>4</i>                         | <i>9</i>  | <i>16</i> | <i>10</i>      | <i>9</i>  | <i>11</i>       | <i>15</i> |
| SDG 10               | 7                                | 5         | 4         | 4              | 4         | 6               | 2         |
| SDG 11               | <i>6</i>                         | <i>6</i>  | <i>9</i>  | <i>13</i>      | <i>13</i> | <i>4</i>        | <i>5</i>  |
| SDG 12               | 12                               | 7         | 5         | 9              | 10        | 16              | 8         |
| SDG 13               | <b>5</b>                         | <b>8</b>  | <b>6</b>  | <b>6</b>       | <b>5</b>  | <b>3</b>        | <b>4</b>  |
| SDG 14               | <b>14</b>                        | <b>13</b> | <b>15</b> | <b>15</b>      | <b>14</b> | <b>10</b>       | <b>7</b>  |
| SDG 15               | 17                               | 17        | 17        | 14             | 16        | 12              | 13        |
| SDG 16               | <b>3</b>                         | <b>3</b>  | <b>7</b>  | <b>3</b>       | <b>2</b>  | <b>1</b>        | <b>3</b>  |
| SDG 17               | <i>1</i>                         | <i>2</i>  | <i>2</i>  | <i>5</i>       | <i>6</i>  | <i>8</i>        | <i>9</i>  |
| Rank Corr ( $\rho$ ) | 1.00                             | 0.85      | 0.60      | 0.63           | 0.71      | 0.54            | 0.36      |

*Adjusted (register-removed, canonical).*

*Notes:* Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. Panel (a) reports the register-removed (adjusted, canonical) ranking; Panel (b) reports the naive baseline (un-adjusted, raw) ranking. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section I.3).

LR = logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix I.4). Rank Corr ( $\rho$ ) is the Spearman correlation of each column’s SDG gap ranks against that panel’s canonical adjusted (Panel a) or raw (Panel b) MPNet-LR column.

**Table 16.** Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

| SDG                  | Encoder (embedding architecture) |           |           |                |           |                 |           |
|----------------------|----------------------------------|-----------|-----------|----------------|-----------|-----------------|-----------|
|                      | MPNet (768-d)                    |           |           | MiniLM (384-d) |           | SciBERT (768-d) |           |
|                      | LR                               | MLP       | ZS        | LR             | MLP       | LR              | MLP       |
| SDG 1                | 5                                | 8         | 7         | 17             | 17        | 16              | 1         |
| SDG 2                | 10                               | 11        | 10        | 14             | 14        | 14              | 14        |
| SDG 3                | <b>1</b>                         | <b>4</b>  | <b>4</b>  | <b>2</b>       | <b>2</b>  | <b>1</b>        | <b>3</b>  |
| SDG 4                | <b>15</b>                        | <b>15</b> | <b>15</b> | <b>15</b>      | <b>15</b> | <b>10</b>       | <b>12</b> |
| SDG 5                | 13                               | 9         | 16        | 4              | 8         | 11              | 15        |
| SDG 6                | <b>12</b>                        | <b>14</b> | <b>5</b>  | <b>13</b>      | <b>16</b> | <b>6</b>        | <b>8</b>  |
| SDG 7                | <b>4</b>                         | <b>7</b>  | <b>1</b>  | <b>5</b>       | <b>4</b>  | <b>4</b>        | <b>4</b>  |
| SDG 8                | 6                                | 3         | 14        | 10             | 12        | 15              | 11        |
| SDG 9                | <b>9</b>                         | <b>12</b> | <b>17</b> | <b>8</b>       | <b>6</b>  | <b>7</b>        | <b>13</b> |
| SDG 10               | 14                               | 13        | 12        | 3              | 3         | 5               | 7         |
| SDG 11               | 3                                | 2         | 2         | 6              | 7         | 3               | 2         |
| SDG 12               | 11                               | 5         | 9         | 7              | 5         | 13              | 5         |
| SDG 13               | <b>2</b>                         | <b>1</b>  | <b>6</b>  | <b>1</b>       | <b>1</b>  | <b>2</b>        | <b>6</b>  |
| SDG 14               | 7                                | 6         | 3         | 11             | 10        | 8               | 9         |
| SDG 15               | 16                               | 17        | 11        | 9              | 13        | 9               | 10        |
| SDG 16               | 8                                | 10        | 13        | 12             | 11        | 12              | 16        |
| SDG 17               | <b>17</b>                        | <b>16</b> | <b>8</b>  | <b>16</b>      | <b>9</b>  | <b>17</b>       | <b>17</b> |
| Rank Corr ( $\rho$ ) | 1.00                             | 0.86      | 0.53      | 0.40           | 0.37      | 0.46            | 0.62      |

*Raw (naive baseline, un-adjusted).*

*Notes:* Each cell reports the within-SDG semantic-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. Panel (a) reports the register-removed (adjusted, canonical) ranking; Panel (b) reports the naive baseline (un-adjusted, raw) ranking. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section I.3).

LR = logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix I.4). Rank Corr ( $\rho$ ) is the Spearman correlation of each column’s SDG gap ranks against that panel’s canonical adjusted (Panel a) or raw (Panel b) MPNet-LR column.

**Table 17.** Domain-encoder sensitivity of within-SDG coverage-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

| SDG                  | Encoder (embedding architecture) |           |           |                |           |                 |           |
|----------------------|----------------------------------|-----------|-----------|----------------|-----------|-----------------|-----------|
|                      | MPNet (768-d)                    |           |           | MiniLM (384-d) |           | SciBERT (768-d) |           |
|                      | LR                               | MLP       | ZS        | LR             | MLP       | LR              | MLP       |
| SDG 1                | <b>7</b>                         | <b>6</b>  | <b>7</b>  | <b>7</b>       | <b>7</b>  | <b>7</b>        | <b>6</b>  |
| SDG 2                | <i>15</i>                        | <i>13</i> | <i>14</i> | <i>11</i>      | <i>11</i> | <i>11</i>       | <i>9</i>  |
| SDG 3                | <b>1</b>                         | <b>1</b>  | <b>2</b>  | <b>1</b>       | <b>1</b>  | <b>1</b>        | <b>1</b>  |
| SDG 4                | <b>4</b>                         | <b>4</b>  | <b>3</b>  | <b>5</b>       | <b>4</b>  | <b>3</b>        | <b>3</b>  |
| SDG 5                | <i>8</i>                         | <i>10</i> | <i>8</i>  | <i>12</i>      | <i>9</i>  | <i>8</i>        | <i>8</i>  |
| SDG 6                | <b>9</b>                         | <b>8</b>  | <b>12</b> | <b>8</b>       | <b>10</b> | <b>12</b>       | <b>10</b> |
| SDG 7                | <b>16</b>                        | <b>17</b> | <b>6</b>  | <b>16</b>      | <b>17</b> | <b>17</b>       | <b>17</b> |
| SDG 8                | 11                               | 15        | 9         | 9              | 8         | 9               | 11        |
| SDG 9                | <b>2</b>                         | <b>2</b>  | <b>4</b>  | <b>2</b>       | <b>2</b>  | <b>2</b>        | <b>4</b>  |
| SDG 10               | <b>14</b>                        | <b>9</b>  | <b>11</b> | <b>15</b>      | <b>14</b> | <b>15</b>       | <b>14</b> |
| SDG 11               | <i>17</i>                        | <i>16</i> | <i>15</i> | <i>13</i>      | <i>13</i> | <i>14</i>       | <i>13</i> |
| SDG 12               | 12                               | 14        | 13        | 14             | 15        | 13              | 15        |
| SDG 13               | <b>5</b>                         | <b>5</b>  | <b>5</b>  | <b>6</b>       | <b>6</b>  | <b>6</b>        | <b>5</b>  |
| SDG 14               | <b>10</b>                        | <b>12</b> | <b>16</b> | <b>10</b>      | <b>12</b> | <b>10</b>       | <b>12</b> |
| SDG 15               | <i>13</i>                        | <i>11</i> | <i>17</i> | <i>17</i>      | <i>16</i> | <i>16</i>       | <i>16</i> |
| SDG 16               | 6                                | 7         | 10        | 3              | 3         | 5               | 7         |
| SDG 17               | <b>3</b>                         | <b>3</b>  | <b>1</b>  | <b>4</b>       | <b>5</b>  | <b>4</b>        | <b>2</b>  |
| Rank Corr ( $\rho$ ) | 1.00                             | 0.92      | 0.75      | 0.89           | 0.90      | 0.93            | 0.90      |

*Notes:* Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section I.3). LR = canonical supervised logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only. Coverage gap is document-weighted (Assumption A19) for all methods.

Rank Corr ( $\rho$ ) is the Spearman correlation of each column’s SDG coverage-gap ranks against the canonical MPNet-LR column.

**Table 18.** Cross-sensitivity robustness of within-SDG coverage-gap rankings across policy source and retrieval strategy.

| SDG                  | Policy source |         |      |       | Retrieval |      |
|----------------------|---------------|---------|------|-------|-----------|------|
|                      | Canon         | Curated | SDGi | UNGDC | LR        | MLP  |
| SDG 1                | 7             | 17      | 4    | 11    | 6         | 6    |
| SDG 2                | 15            | 8       | 12   | 12    | 12        | 13   |
| SDG 3                | 1             | 3       | 1    | 2     | 2         | 2    |
| SDG 4                | 4             | 4       | 3    | 6     | 7         | 14   |
| SDG 5                | 8             | 11      | 7    | 17    | 8         | 10   |
| SDG 6                | 9             | 15      | 9    | 16    | 9         | 8    |
| SDG 7                | 16            | 7       | 17   | 8     | 17        | 17   |
| SDG 8                | 11            | 9       | 8    | 9     | 15        | 15   |
| SDG 9                | 2             | 1       | 2    | 4     | 1         | 1    |
| SDG 10               | 14            | 16      | 13   | 15    | 14        | 9    |
| SDG 11               | 17            | 5       | 15   | 7     | 10        | 7    |
| SDG 12               | 12            | 14      | 10   | 10    | 16        | 16   |
| SDG 13               | 5             | 6       | 16   | 5     | 5         | 5    |
| SDG 14               | 10            | 13      | 14   | 13    | 11        | 11   |
| SDG 15               | 13            | 10      | 11   | 14    | 13        | 12   |
| SDG 16               | 6             | 12      | 6    | 1     | 4         | 4    |
| SDG 17               | 3             | 2       | 5    | 3     | 3         | 3    |
| Rank Corr ( $\rho$ ) | 1.00          | 0.40    | 0.77 | 0.54  | 0.87      | 0.66 |

*Notes:* Each cell reports the within-SDG coverage-gap rank (1 = largest gap, 17 = smallest gap). The Canon column is the canonical MPNet—LR ranking. Coverage gap =  $|\text{research proportion} - \text{policy proportion}|$  per SDG, using document-weighted policy proportions (Assumption A19). Policy-source columns compare the keyword-retrieved research profile against each policy-source family. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. The Segment-cap axis is omitted: coverage gap is segment-cap-independent. Rank Corr ( $\rho$ ) is the Spearman correlation of each column’s SDG coverage-gap ranks against the canonical column.

**Table 19.** Coverage gap values across embedding architectures, assignment methods, and retrieval strategies.

| SDG | Coverage gap (%) |      |      |        |      |         |      |         |
|-----|------------------|------|------|--------|------|---------|------|---------|
|     | MPNet            |      |      | MiniLM |      | SciBERT |      | Concept |
|     | LR               | MLP  | ZS   | LR     | MLP  | LR      | MLP  |         |
| 1   | 4.1              | 4.5  | 3.8  | 4.1    | 4.1  | 5.1     | 5.6  | 3.9     |
| 2   | 1.9              | 1.5  | 0.7  | 2.1    | 2.3  | 2.6     | 2.9  | 2.5     |
| 3   | 21.0             | 16.8 | 17.2 | 22.0   | 22.4 | 22.3    | 23.2 | 21.0    |
| 4   | 10.8             | 8.9  | 11.7 | 8.9    | 8.8  | 12.8    | 12.3 | 3.5     |
| 5   | 3.1              | 2.2  | 2.6  | 1.7    | 2.4  | 3.9     | 3.4  | 3.3     |
| 6   | 2.7              | 2.6  | 1.1  | 2.3    | 2.3  | 2.5     | 2.4  | 2.9     |
| 7   | 0.2              | 0.4  | 4.0  | 1.0    | 0.8  | 0.6     | 0.1  | 0.4     |
| 8   | 2.1              | 1.1  | 1.9  | 2.3    | 2.5  | 3.7     | 2.4  | 2.0     |
| 9   | 15.8             | 15.0 | 9.4  | 12.5   | 15.1 | 17.6    | 11.6 | 21.6    |
| 10  | 2.0              | 2.2  | 1.3  | 1.1    | 1.6  | 1.6     | 1.5  | 2.1     |
| 11  | 0.2              | 0.5  | 0.3  | 1.6    | 1.6  | 1.8     | 1.6  | 2.7     |
| 12  | 2.1              | 1.1  | 1.0  | 1.6    | 1.6  | 2.5     | 1.0  | 0.9     |
| 13  | 7.6              | 5.7  | 8.5  | 5.1    | 5.9  | 5.9     | 8.6  | 7.1     |
| 14  | 2.2              | 1.9  | 0.2  | 2.2    | 2.2  | 2.8     | 2.4  | 2.5     |
| 15  | 2.0              | 1.9  | 0.1  | 0.1    | 1.0  | 1.3     | 0.4  | 2.1     |
| 16  | 6.1              | 3.9  | 1.7  | 11.2   | 11.1 | 9.1     | 4.8  | 8.5     |
| 17  | 11.4             | 12.1 | 20.2 | 8.9    | 8.5  | 9.1     | 13.2 | 11.5    |

*Notes:* Coverage gap = |research proportion – policy proportion| per SDG in percentage points, using document-weighted policy proportions (Assumption A19) for all methods.

**Table 20.** (continued) Semantic gap values across embedding architectures, assignment methods, and retrieval strategies.

| SDG | Semantic gap (cosine) |       |       |        |       |         |       | Concept |
|-----|-----------------------|-------|-------|--------|-------|---------|-------|---------|
|     | MPNet                 |       |       | MiniLM |       | SciBERT |       |         |
|     | LR                    | MLP   | ZS    | LR     | MLP   | LR      | MLP   |         |
| 1   | 0.415                 | 0.413 | 0.407 | 0.222  | 0.235 | 0.032   | 0.061 | 0.827   |
| 2   | 0.299                 | 0.330 | 0.345 | 0.297  | 0.306 | 0.037   | 0.041 | 0.428   |
| 3   | 0.494                 | 0.480 | 0.436 | 0.495  | 0.507 | 0.056   | 0.056 | 0.663   |
| 4   | 0.267                 | 0.267 | 0.265 | 0.291  | 0.298 | 0.044   | 0.043 | 0.402   |
| 5   | 0.278                 | 0.380 | 0.246 | 0.470  | 0.360 | 0.041   | 0.040 | 0.575   |
| 6   | 0.292                 | 0.292 | 0.428 | 0.298  | 0.294 | 0.045   | 0.046 | 0.384   |
| 7   | 0.428                 | 0.429 | 0.525 | 0.430  | 0.414 | 0.052   | 0.053 | 0.514   |
| 8   | 0.387                 | 0.507 | 0.278 | 0.336  | 0.317 | 0.033   | 0.045 | 0.625   |
| 9   | 0.314                 | 0.327 | 0.243 | 0.350  | 0.384 | 0.045   | 0.041 | 0.421   |
| 10  | 0.271                 | 0.317 | 0.305 | 0.485  | 0.434 | 0.047   | 0.049 | 0.458   |
| 11  | 0.449                 | 0.510 | 0.477 | 0.397  | 0.371 | 0.052   | 0.057 | 0.693   |
| 12  | 0.296                 | 0.438 | 0.369 | 0.391  | 0.411 | 0.039   | 0.052 | 0.400   |
| 13  | 0.480                 | 0.525 | 0.423 | 0.541  | 0.526 | 0.053   | 0.051 | 0.598   |
| 14  | 0.358                 | 0.431 | 0.452 | 0.330  | 0.334 | 0.044   | 0.046 | 0.504   |
| 15  | 0.239                 | 0.231 | 0.325 | 0.345  | 0.307 | 0.044   | 0.045 | 0.346   |
| 16  | 0.329                 | 0.332 | 0.294 | 0.322  | 0.322 | 0.040   | 0.037 | 0.541   |
| 17  | 0.177                 | 0.267 | 0.398 | 0.241  | 0.355 | 0.023   | 0.030 | 0.312   |

*Notes:* Semantic gap =  $1 - \cos(\mathbf{r}_{\text{SDG}}, \mathbf{p}_{\text{SDG}})$  where  $\mathbf{r}_{\text{SDG}}$  and  $\mathbf{p}_{\text{SDG}}$  are the research and policy centroids; policy centroids use a 50-segment-per-document cap. All values are cosine units. This is a *values* table reporting the adjusted (register-removed) semantic gap only; it carries no raw/adjusted panel split, which is shown in the rank-robustness tables (Section I.2).

**Table 21.** Assignment agreement between the supervised LR classifier and the zero-shot nearest-centroid rule, by SDG and overall, under the canonical MPNet encoder and keyword retrieval.

| SDG                           | Research (segments) |         |           | Policy (segments) |         | Semantic-gap rank |    |            |
|-------------------------------|---------------------|---------|-----------|-------------------|---------|-------------------|----|------------|
|                               | $n_{LR}$            | Agree % | Res MLP % | $n_{LR}$          | Agree % | LR                | ZS | $ \Delta $ |
| SDG 1 (No Poverty)            | 13,459              | 30.9    | 14.9      | 1,700             | 70.8    | 5                 | 7  | 2          |
| SDG 2 (Zero Hunger)           | 78,591              | 77.6    | 59.8      | 1,723             | 94.1    | 10                | 10 | 0          |
| SDG 3 (Good Health)           | 859,355             | 75.0    | 79.8      | 2,934             | 87.7    | 1                 | 4  | 3          |
| SDG 4 (Quality Education)     | 470,002             | 82.4    | 82.6      | 2,055             | 91.3    | 15                | 15 | 0          |
| SDG 5 (Gender Equality)       | 64,940              | 56.3    | 37.0      | 2,026             | 88.5    | 13                | 16 | 3          |
| SDG 6 (Clean Water)           | 35,137              | 85.3    | 75.2      | 1,206             | 94.6    | 12                | 5  | 7          |
| SDG 7 (Clean Energy)          | 119,467             | 88.3    | 81.9      | 1,808             | 84.1    | 4                 | 1  | 3          |
| SDG 8 (Decent Work)           | 65,500              | 44.4    | 20.7      | 2,016             | 71.3    | 6                 | 14 | 8          |
| SDG 9 (Industry & Infra.)     | 577,833             | 50.8    | 45.1      | 4,066             | 81.0    | 9                 | 17 | 8          |
| SDG 10 (Reduced Inequalities) | 30,323              | 23.6    | 12.8      | 1,123             | 56.9    | 14                | 12 | 2          |
| SDG 11 (Sustainable Cities)   | 146,508             | 77.2    | 64.0      | 1,658             | 88.3    | 3                 | 2  | 1          |
| SDG 12 (Responsible Cons.)    | 46,199              | 60.9    | 38.5      | 1,187             | 91.8    | 11                | 9  | 2          |
| SDG 13 (Climate Action)       | 121,557             | 40.5    | 28.4      | 3,399             | 84.5    | 2                 | 6  | 4          |
| SDG 14 (Life Below Water)     | 51,458              | 89.6    | 69.7      | 1,634             | 92.3    | 7                 | 3  | 4          |
| SDG 15 (Life on Land)         | 78,102              | 83.5    | 70.3      | 1,828             | 84.1    | 16                | 11 | 5          |
| SDG 16 (Peace & Justice)      | 337,317             | 55.6    | 46.1      | 5,725             | 65.0    | 8                 | 13 | 5          |
| SDG 17 (Partnerships)         | 9,396               | 26.1    | 11.7      | 4,509             | 84.1    | 17                | 8  | 9          |
| Overall                       | 3,105,144           | 67.3    | 59.5      | 40,597            | 81.5    |                   |    |            |

*Notes:* Agree % is the share of LR-assigned texts that the zero-shot rule also assigns to the same SDG. Semantic-gap ranks are 1 = largest gap;  $|\Delta|$  is the absolute rank difference between the two methods. Assignment agreement is computed over all texts (research: each segment; policy: each segment), without the 50-segment-per-document cap, which applies only to centroid building. The zero-shot rule assigns each text to the SDG with the highest cosine similarity to the train-split reference centroids ( $\mathbf{s}_k$ ); the LR rule uses the trained classifier’s argmax. MLP agreement against zero-shot is reported for both corpora: research MLP-vs-ZS agreement is computed from the persisted per-shard research scores and shown as the overall Res MLP % (59.5%), alongside the policy MLP-vs-ZS rates.



**Table 22.** Pearson  $r$  on the raw coverage and semantic-gap magnitudes (not ranked).

|                                                                                 | Raw gap             | Adj. gap            | Register            |
|---------------------------------------------------------------------------------|---------------------|---------------------|---------------------|
| <b>H1a. Coverage gap <math>\leftrightarrow</math> Semantic gap</b>              |                     |                     |                     |
| MPNet LR                                                                        | +0.102              | +0.694**            | -0.382              |
| MPNet MLP                                                                       | -0.069              | +0.441 <sup>†</sup> | -0.404              |
| MPNet ZS                                                                        | -0.171              | +0.230              | -0.333              |
| MiniLM LR                                                                       | +0.068              | +0.474 <sup>†</sup> | -0.226              |
| MiniLM MLP                                                                      | +0.333              | +0.611**            | -0.093              |
| SciBERT LR                                                                      | +0.129              | +0.152              | -0.080              |
| SciBERT MLP                                                                     | -0.108              | -0.189              | +0.138              |
| Concept LR                                                                      | +0.030              | +0.445 <sup>†</sup> | -0.357              |
| Concept MLP                                                                     | -0.092              | +0.196              | -0.310              |
| <b>H1b. Policy–research dominance <math>\leftrightarrow</math> Semantic gap</b> |                     |                     |                     |
| MPNet LR                                                                        | +0.274              | +0.196              | +0.115              |
| MPNet MLP                                                                       | +0.050              | +0.090              | -0.014              |
| MPNet ZS                                                                        | -0.152              | -0.227              | +0.033              |
| MiniLM LR                                                                       | +0.293              | +0.119              | +0.240              |
| MiniLM MLP                                                                      | +0.353              | +0.150              | +0.260              |
| SciBERT LR                                                                      | +0.460 <sup>†</sup> | -0.156              | +0.420 <sup>†</sup> |
| SciBERT MLP                                                                     | +0.194              | -0.358              | +0.484*             |
| Concept LR                                                                      | +0.117              | +0.120              | +0.021              |
| Concept MLP                                                                     | -0.060              | +0.046              | -0.119              |
| <b>H1c. Research coverage <math>\leftrightarrow</math> Semantic gap</b>         |                     |                     |                     |
| MPNet LR                                                                        | +0.288              | +0.464 <sup>†</sup> | -0.055              |
| MPNet MLP                                                                       | +0.032              | +0.308              | -0.197              |
| MPNet ZS                                                                        | -0.191              | -0.117              | -0.087              |
| MiniLM LR                                                                       | +0.302              | +0.355              | +0.101              |
| MiniLM MLP                                                                      | +0.429 <sup>†</sup> | +0.438 <sup>†</sup> | +0.132              |
| SciBERT LR                                                                      | +0.419 <sup>†</sup> | +0.208              | +0.029              |
| SciBERT MLP                                                                     | +0.029              | -0.234              | +0.262              |
| Concept LR                                                                      | +0.134              | +0.262              | -0.085              |
| Concept MLP                                                                     | -0.074              | +0.113              | -0.205              |
| <b>H1d. Policy coverage <math>\leftrightarrow</math> Semantic gap</b>           |                     |                     |                     |
| MPNet LR                                                                        | -0.018              | +0.501*             | -0.359              |
| MPNet MLP                                                                       | -0.045              | +0.418 <sup>†</sup> | -0.361              |
| MPNet ZS                                                                        | -0.052              | +0.258              | -0.244              |
| MiniLM LR                                                                       | -0.041              | +0.381              | -0.284              |
| MiniLM MLP                                                                      | +0.064              | +0.466*             | -0.269              |
| SciBERT LR                                                                      | -0.138              | +0.728**            | -0.813***           |
| SciBERT MLP                                                                     | -0.344              | +0.287              | -0.492*             |
| Concept LR                                                                      | +0.009              | +0.269              | -0.225              |
| Concept MLP                                                                     | -0.017              | +0.130              | -0.154              |

*Notes:* Same H1a–H1d grid and column layout as Table 3; significance stars use the two-sided Monte Carlo permutation  $p$ -value (\*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , \*  $p < 0.05$ , <sup>†</sup>  $p < 0.10$ ). Identical inputs and configuration grid to Table 3; only the correlation function changes from Spearman  $\rho$  (ranks) to Pearson  $r$  (raw magnitudes). ZS is MPNet-only per manuscript scope.

**Table 23.** Per-SDG policy source-family sensitivity: coverage share and semantic gap.

| SDG    | Full Corpus  |              | Curated AI/SDG |              | SDGi VNR/VLR |              | UNGDC      |              |
|--------|--------------|--------------|----------------|--------------|--------------|--------------|------------|--------------|
|        | pol.% (n)    | sem. (n)     | pol.% (n)      | sem. (n)     | pol.% (n)    | sem. (n)     | pol.% (n)  | sem. (n)     |
| 1      | 4.5 (288)    | 0.42 (1,689) | 1.1 (1)        | 0.44 (357)   | 5.9 (250)    | 0.42 (1,191) | 1.8 (37)   | 0.56 (141)   |
| 2      | 4.3 (274)    | 0.30 (1,508) | 0.0 (0)        | 0.35 (240)   | 5.8 (246)    | 0.29 (1,196) | 1.4 (28)   | 0.45 (72)    |
| 3      | 4.7 (298)    | 0.49 (2,408) | 6.4 (6)        | 0.36 (643)   | 6.7 (284)    | 0.55 (1,707) | 0.4 (8)    | 0.70 (58)    |
| 4      | 4.8 (305)    | 0.27 (2,055) | 0.0 (0)        | 0.21 (274)   | 7.0 (296)    | 0.29 (1,718) | 0.4 (9)    | 0.40 (63)    |
| 5      | 5.1 (324)    | 0.28 (2,005) | 0.0 (0)        | 0.28 (272)   | 6.6 (278)    | 0.28 (1,531) | 2.2 (46)   | 0.41 (202)   |
| 6      | 3.7 (238)    | 0.29 (1,206) | 0.0 (0)        | 0.35 (137)   | 5.3 (226)    | 0.29 (1,038) | 0.6 (12)   | 0.42 (31)    |
| 7      | 4.3 (273)    | 0.43 (1,578) | 1.1 (1)        | 0.51 (358)   | 6.0 (253)    | 0.41 (1,147) | 0.9 (19)   | 0.59 (73)    |
| 8      | 4.3 (276)    | 0.39 (1,886) | 0.0 (0)        | 0.40 (309)   | 6.5 (275)    | 0.40 (1,558) | 0.0 (1)    | 0.54 (19)    |
| 9      | 4.3 (274)    | 0.31 (2,986) | 50.0 (47)      | 0.27 (1,838) | 5.1 (215)    | 0.54 (1,116) | 0.6 (12)   | 0.46 (32)    |
| 10     | 2.9 (182)    | 0.27 (1,085) | 0.0 (0)        | 0.33 (258)   | 4.2 (176)    | 0.28 (774)   | 0.3 (6)    | 0.40 (53)    |
| 11     | 5.3 (338)    | 0.45 (1,658) | 0.0 (0)        | 0.43 (175)   | 8.0 (336)    | 0.46 (1,474) | 0.1 (2)    | 0.57 (9)     |
| 12     | 3.5 (226)    | 0.30 (1,132) | 0.0 (0)        | 0.38 (178)   | 5.3 (225)    | 0.30 (947)   | 0.0 (1)    | 0.49 (7)     |
| 13     | 11.3 (718)   | 0.48 (3,084) | 7.4 (7)        | 0.44 (613)   | 6.0 (252)    | 0.44 (1,268) | 22.4 (459) | 0.58 (1,203) |
| 14     | 3.8 (241)    | 0.36 (1,293) | 0.0 (0)        | 0.37 (216)   | 4.5 (189)    | 0.35 (856)   | 2.5 (52)   | 0.49 (221)   |
| 15     | 4.2 (266)    | 0.24 (1,484) | 0.0 (0)        | 0.29 (299)   | 5.7 (239)    | 0.24 (1,108) | 1.3 (27)   | 0.41 (77)    |
| 16     | 17.3 (1,100) | 0.33 (4,890) | 12.8 (12)      | 0.20 (1,092) | 5.9 (250)    | 0.38 (1,395) | 40.9 (838) | 0.51 (2,403) |
| 17     | 11.7 (746)   | 0.18 (3,645) | 21.3 (20)      | 0.18 (859)   | 5.6 (235)    | 0.16 (1,322) | 24.0 (491) | 0.26 (1,464) |
| $\rho$ | 1.00         | 1.00         | 0.36           | 0.73         | 0.61         | 0.91         | 0.45       | 0.97         |

*Notes:* For each policy source family, **pol.% (n)** is the document-weighted policy share (%) with the document count in parentheses; **sem. (n)** is the within-SDG research-policy semantic gap with the capped segment count in parentheses. Read across a row to see whether the gap is stable or driven by one family.

**Table 24.** Per-family interaction test: coverage predictors vs the within-SDG semantic gap.

| Family         | $n$ | Research share ( $\rho, p$ ) | Coverage gap ( $\rho, p$ ) | Research share excl. SDG 4 ( $\rho, p$ ) |
|----------------|-----|------------------------------|----------------------------|------------------------------------------|
| Full corpus    | 17  | 0.45 (0.071)                 | -0.01 (0.959)              | 0.61 (0.015)                             |
| Curated AI/SDG | 17  | -0.06 (0.833)                | -0.22 (0.386)              | 0.05 (0.848)                             |
| SDGi VNR/VLR   | 17  | 0.60 (0.014)                 | 0.04 (0.873)               | 0.70 (0.003)                             |
| UNGDC speeches | 15  | 0.42 (0.115)                 | 0.35 (0.195)               | 0.61 (0.024)                             |

*Notes:* Spearman  $\rho$  (with two-sided Monte Carlo permutation  $p$ , 100,000 resamples, fixed seed) between each coverage predictor and the within-SDG semantic gap across the reliable SDGs ( $n$ ). Research share is the research-corpus proportion per SDG (family-independent); the coverage gap is  $|\text{research} - \text{policy}|$  using the family’s document-weighted policy share. The curated AI/SDG family is the symmetric AI/SDG-vs-AI/SDG control matching the research corpus’s  $\text{AI} \cap \text{SDG}$  scope.

## J Pooled Regression: Coverage Predictors and the Semantic Gap

This appendix pools all 24 encoder–retrieval–method–cap configurations into a single panel ( $N = 408$ ) and regresses the within-SDG semantic gap on coverage predictors using OLS with cluster-robust standard errors by SDG (17 clusters). The dependent variable is the rank of the semantic gap within each configuration (1–17 per config, ties averaged); predictors are expressed in percentage points (0–100). Table 25 reports the full 23-specification grid across core regressions (Panel A), robustness checks (Panel B), interaction terms (Panel C), and alternative functional forms (Panel D).

The adjusted semantic gap is positively associated with the coverage gap ( $b = +0.18$ ,  $p = 0.069$ ; Panel A, column 1); the coefficient is marginally significant at conventional thresholds but is confirmed by the supervised-only subsample ( $p = 0.040$ ), the MPNet-only subsample ( $p = 0.003$ ), and the raw-DV specification in cosine units ( $p = 0.007$ ). The raw gap and the register component show no detectable association with any coverage predictor, consistent with the cancellation pattern reported in Section 4.3. Policy coverage (the share of policy documents assigned to a given SDG) is a stronger predictor of the adjusted gap ( $b = +0.41$ ,  $p = 0.003$ ), capturing how concentrated policy attention is on each goal. The coverage-gap-by-encoder interaction (Panel C) reveals that the effect is strongest for MPNet and reverses for SciBERT, suggesting that domain-specialised pretraining compresses the adjusted semantic gap regardless of coverage structure.

**Table 25.** Pooled OLS: coverage predictors and the within-SDG semantic gap rank across 24 configurations.

|  | Class |  |  |     |  |  |     |  |  |     |  |  | Substructure |  |  |     |  |  |     |  |  |     |  |  | Interstructure |  |  |      |  |  |      |  |  |      |  |  | Functional form |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |      |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       |  |  |       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 | (76) |  |  | (77) |  |  | (78) |  |  | (79) |  |  | (80) |  |  | (81) |  |  | (82) |  |  | (83) |  |  | (84) |  |  | (85) |  |  | (86) |  |  | (87) |  |  | (88) |  |  | (89) |  |  | (90) |  |  | (91) |  |  | (92) |  |  | (93) |  |  | (94) |  |  | (95) |  |  | (96) |  |  | (97) |  |  | (98) |  |  | (99) |  |  | (100) |  |  | (101) |  |  | (102) |  |  | (103) |  |  | (104) |  |  | (105) |  |  | (106) |  |  | (107) |  |  | (108) |  |  | (109) |  |  | (110) |  |  | (111) |  |  | (112) |  |  | (113) |  |  | (114) |  |  | (115) |  |  | (116) |  |  | (117) |  |  | (118) |  |  | (119) |  |  | (120) |  |  | (121) |  |  | (122) |  |  | (123) |  |  | (124) |  |  | (125) |  |  | (126) |  |  | (127) |  |  | (128) |  |  | (129) |  |  | (130) |  |  | (131) |  |  | (132) |  |  | (133) |  |  | (134) |  |  | (135) |  |  | (136) |  |  | (137) |  |  | (138) |  |  | (139) |  |  | (140) |  |  | (141) |  |  | (142) |  |  | (143) |  |  | (144) |  |  | (145) |  |  | (146) |  |  | (147) |  |  | (148) |  |  | (149) |  |  | (150) |  |  | (151) |  |  | (152) |  |  | (153) |  |  | (154) |  |  | (155) |  |  | (156) |  |  | (157) |  |  | (158) |  |  | (159) |  |  | (160) |  |  | (161) |  |  | (162) |  |  | (163) |  |  | (164) |  |  | (165) |  |  | (166) |  |  | (167) |  |  | (168) |  |  | (169) |  |  | (170) |  |  | (171) |  |  | (172) |  |  | (173) |  |  | (174) |  |  | (175) |  |  | (176) |  |  | (177) |  |  | (178) |  |  | (179) |  |  | (180) |  |  | (181) |  |  | (182) |  |  | (183) |  |  | (184) |  |  | (185) |  |  | (186) |  |  | (187) |  |  | (188) |  |  | (189) |  |  | (190) |  |  | (191) |  |  | (192) |  |  | (193) |  |  | (194) |  |  | (195) |  |  | (196) |  |  | (197) |  |  | (198) |  |  | (199) |  |  | (200) |  |  | (201) |  |  | (202) |  |  | (203) |  |  | (204) |  |  | (205) |  |  | (206) |  |  | (207) |  |  | (208) |  |  | (209) |  |  | (210) |  |  | (211) |  |  | (212) |  |  | (213) |  |  | (214) |  |  | (215) |  |  | (216) |  |  | (217) |  |  | (218) |  |  | (219) |  |  | (220) |  |  | (221) |  |  | (222) |  |  | (223) |  |  | (224) |  |  | (225) |  |  | (226) |  |  | (227) |  |  | (228) |  |  | (229) |  |  | (230) |  |  | (231) |  |  | (232) |  |  | (233) |  |  | (234) |  |  | (235) |  |  | (236) |  |  | (237) |  |  | (238) |  |  | (239) |  |  | (240) |  |  | (241) |  |  | (242) |  |  | (243) |  |  | (244) |  |  | (245) |  |  | (246) |  |  | (247) |  |  | (248) |  |  | (249) |  |  | (250) |  |  | (251) |  |  | (252) |  |  | (253) |  |  | (254) |  |  | (255) |  |  | (256) |  |  | (257) |  |  | (258) |  |  | (259) |  |  | (260) |  |  | (261) |  |  | (262) |  |  | (263) |  |  | (264) |  |  | (265) |  |  | (266) |  |  | (267) |  |  | (268) |  |  | (269) |  |  | (270) |  |  | (271) |  |  | (272) |  |  | (273) |  |  | (274) |  |  | (275) |  |  | (276) |  |  | (277) |  |  | (278) |  |  | (279) |  |  | (280) |  |  | (281) |  |  | (282) |  |  | (283) |  |  | (284) |  |  | (285) |  |  | (286) |  |  | (287) |  |  | (288) |  |  | (289) |  |  | (290) |  |  | (291) |  |  | (292) |  |  | (293) |  |  | (294) |  |  | (295) |  |  | (296) |  |  | (297) |  |  | (298) |  |  | (299) |  |  | (300) |  |  | (301) |  |  | (302) |  |  | (303) |  |  | (304) |  |  | (305) |  |  | (306) |  |  | (307) |  |  | (308) |  |  | (309) |  |  | (310) |  |  | (311) |  |  | (312) |  |  | (313) |  |  | (314) |  |  | (315) |  |  | (316) |  |  | (317) |  |  | (318) |  |  | (319) |  |  | (320) |  |  | (321) |  |  | (322) |  |  | (323) |  |  | (324) |  |  | (325) |  |  | (326) |  |  | (327) |  |  | (328) |  |  | (329) |  |  | (330) |  |  | (331) |  |  | (332) |  |  | (333) |  |  | (334) |  |  | (335) |  |  | (336) |  |  | (337) |  |  | (338) |  |  | (339) |  |  | (340) |  |  | (341) |  |  | (342) |  |  | (343) |  |  | (344) |  |  | (345) |  |  | (346) |  |  | (347) |  |  | (348) |  |  | (349) |  |  | (350) |  |  | (351) |  |  | (352) |  |  | (353) |  |  | (354) |  |  | (355) |  |  | (356) |  |  | (357) |  |  | (358) |  |  | (359) |  |  | (360) |  |  | (361) |  |  | (362) |  |  | (363) |  |  | (364) |  |  | (365) |  |  | (366) |  |  | (367) |  |  | (368) |  |  | (369) |  |  | (370) |  |  | (371) |  |  | (372) |  |  | (373) |  |  | (374) |  |  | (375) |  |  | (376) |  |  | (377) |  |  | (378) |  |  | (379) |  |  | (380) |  |  | (381) |  |  | (382) |  |  | (383) |  |  | (384) |  |  | (385) |  |  | (386) |  |  | (387) |  |  | (388) |  |  | (389) |  |  | (390) |  |  | (391) |  |  | (392) |  |  | (393) |  |  | (394) |  |  | (395) |  |  | (396) |  |  | (397) |  |  | (398) |  |  | (399) |  |  | (400) |  |  | (401) |  |  | (402) |  |  | (403) |  |  | (404) |  |  | (405) |  |  | (406) |  |  | (407) |  |  | (408) |  |  | (409) |  |  | (410) |  |  | (411) |  |  | (412) |  |  | (413) |  |  | (414) |  |  | (415) |  |  | (416) |  |  | (417) |  |  | (418) |  |  | (419) |  |  | (420) |  |  | (421) |  |  | (422) |  |  | (423) |  |  | (424) |  |  | (425) |  |  | (426) |  |  | (427) |  |  | (428) |  |  | (429) |  |  | (430) |  |  | (431) |  |  | (432) |  |  | (433) |  |  | (434) |  |  | (435) |  |  | (436) |  |  | (437) |  |  | (438) |  |  | (439) |  |  | (440) |  |  | (441) |  |  | (442) |  |  | (443) |  |  | (444) |  |  | (445) |  |  | (446) |  |  | (447) |  |  | (448) |  |  | (449) |  |  | (450) |  |  | (451) |  |  | (452) |  |  | (453) |  |  | (454) |  |  | (455) |  |  | (456) |  |  | (457) |  |  | (458) |  |  | (459) |  |  | (460) |  |  | (461) |  |  | (462) |  |  | (463) |  |  | (464) |  |  | (465) |  |  | (466) |  |  | (467) |  |  | (468) |  |  | (469) |  |  | (470) |  |  | (471) |  |  | (472) |  |  | (473) |  |  | (474) |  |  | (475) |  |  | (476) |  |  | (477) |  |  | (478) |  |  | (479) |  |  | (480) |  |  | (481) |  |  | (482) |  |  | (483) |  |  | (484) |  |  | (485) |  |  | (486) |  |  | (487) |  |  | (488) |  |  | (489) |  |  | (490) |  |  | (491) |  |  | (492) |  |  | (493) |  |  | (494) |  |  | (495) |  |  | (496) |  |  | (497) |  |  | (498) |  |  | (499) |  |  | (500) |  |  | (501) |  |  | (502) |  |  | (503) |  |  | (504) |  |  | (505) |  |  | (506) |  |  | (507) |  |  | (508) |  |  | (509) |  |  | (510) |  |  | (511) |  |  | (512) |  |  | (513) |  |  | (514) |  |  | (515) |  |  | (516) |  |  | (517) |  |  | (518) |  |  | (519) |  |  | (520) |  |  | (521) |  |  | (522) |  |  | (523) |  |  | (524) |  |  | (525) |  |  | (526) |  |  | (527) |  |  | (528) |  |  | (529) |  |  | (530) |  |  | (531) |  |  | (532) |  |  | (533) |  |  | (534) |  |  | (535) |  |  | (536) |  |  | (537) |  |  | (538) |  |  | (539) |  |  | (540) |  |  | (541) |  |  | (542) |  |  | (543) |  |  | (544) |  |  | (545) |  |  | (546) |  |  | (547) |  |  | (548) |  |  | (549) |  |  | (550) |  |  | (551) |  |  | (552) |  |  | (553) |  |  | (554) |  |  | (555) |  |  | (556) |  |  | (557) |  |  | (558) |  |  | (559) |  |  | (560) |  |  | (561) |  |  | (562) |  |  | (563) |  |  | (564) |  |  | (565) |  |  | (566) |  |  | (567) |  |  | (568) |  |  | (569) |  |  | (570) |  |  | (571) |  |  | (572) |  |  | (573) |  |  | (574) |  |  | (575) |  |  | (576) |  |  | (577) |  |  | (578) |  |  | (579) |  |  | (580) |  |  | (581) |  |  | (582) |  |  | (583) |  |  | (584) |  |  | (585) |  |  | (586) |  |  | (587) |  |  | (588) |  |  | (589) |  |  | (590) |  |  | (591) |  |  | (592) |  |  | (593) |  |  | (594) |  |  | (595) |  |  | (596) |  |  | (597) |  |  | (598) |  |  | (599) |  |  | (600) |  |  | (601) |  |  | (602) |  |  | (603) |  |  | (604) |  |  | (605) |  |  | (606) |  |  | (607) |  |  | (608) |  |  | (609) |  |  | (610) |  |  | (611) |  |  | (612) |  |  | (613) |  |  | (614) |  |  | (615) |  |  | (616) |  |  | (617) |  |  | (618) |  |  | (619) |  |  | (620) |  |  | (621) |  |  | (622) |  |  | (623) |  |  | (624) |  |  | (625) |  |  | (626) |  |  | (627) |  |  | (628) |  |  | (629) |  |  | (630) |  |  | (631) |  |  | (632) |  |  | (633) |  |  | (634) |  |  | (635) |  |  | (636) |  |  | (637) |  |  | (638) |  |  | (639) |  |  | (640) |  |  | (641) |  |  | (642) |  |  | (643) |  |  | (644) |  |  | (645) |  |  | (646) |  |  | (647) |  |  | (648) |  |  | (649) |  |  | (650) |  |  | (651) |  |  | (652) |  |  | (653) |  |  | (654) |  |  | (655) |  |  | (656) |  |  | (657) |  |  | (658) |  |  | (659) |  |  | (660) |  |  | (661) |  |  | (662) |  |  | (663) |  |  | (664) |  |  | (665) |  |  | (666) |  |  | (667) |  |  | (668) |  |  | (669) |  |  | (670) |  |  | (671) |  |  | (672) |  |  | (673) |  |  | (674) |  |  | (675) |  |  | (676) |  |  | (677) |  |  | (678) |  |  | (679) |  |  | (680) |  |  | (681) |  |  | (682) |  |  | (683) |  |  | (684) |  |  | (685) |  |  | (686) |  |  | (687) |  |  | (688) |  |  | (689) |  |  | (690) |  |  | (691) |  |  | (692) |  |  | (693) |  |  | (694) |  |  | (695) |  |  | (696) |  |  | (697) |  |  | (698) |  |  | (699) |  |  | (700) |  |  | (701) |  |  | (702) |  |  | (703) |  |  | (704) |  |  | (705) |  |  | (706) |  |  | (707) |  |  | (708) |  |  | (709) |  |  | (710) |  |  | (711) |  |  | (712) |  |  | (713) |  |  | (714) |  |  | (715) |  |  | (716) |  |  | (717) |  |  | (718) |  |  | (719) |  |  | (720) |  |  | (721) |  |  | (722) |  |  | (723) |  |  | (724) |  |  | (725) |  |  | (726) |  |  | (727) |  |  | (728) |  |  | (729) |  |  | (730) |  |  | (731) |  |  | (732) |  |  | (733) |  |  | (734) |  |  | (735) |  |  | (736) |  |  | (737) |  |  | (738) |  |  | (739) |  |  | (740) |  |  | (741) |  |  | (742) |  |  | (743) |  |  | (744) |  |  | (745) |  |  | (746) |  |  | (747) |  |  | (748) |  |  | (749) |  |  | (750) |  |  | (751) |  |  | (752) |  |  | (753) |  |  | (754) |  |  | (755) |  |  | (756) |  |  | (757) |  |  | (758) |  |  | (759) |  |  | (760) |  |  | (761) |  |  | (762) |  |  | (763) |  |  | (764) |  |  | (765) |  |  | (766) |  |  | (767) |  |  | (768) |  |  | (769) |  |  | (770) |  |  | (771) |  |  | (772) |  |  | (773) |  |  | (774) |  |  | (775) |  |  | (776) |  |  | (777) |  |  | (778) |  |  | (779) |  |  | (780) |  |  | (781) |  |  | (782) |  |  | (783) |  |  | (784) |  |  | (785) |  |  | (786) |  |  | (787) |  |  | (788) |  |  | (789) |  |  | (790) |  |  | (791) |  |  | (792) |  |  | (793) |  |  | (794) |  |  | (795) |  |  | (796) |  |  | (797) |  |  | (798) |  |  | (799) |  |  | (800) |  |  | (801) |  |  | (802) |  |  | (803) |  |  | (804) |  |  | (805) |  |  | (806) |  |  | (807) |  |  | (808) |  |  | (809) |  |  | (810) |  |  | (811) |  |  | (812) |  |  | (813) |  |  | (814) |  |  | (815) |  |  | (816) |  |  | (817) |  |  | (818) |  |  | (819) |  |  | (820) |  |  | (821) |  |  | (822) |  |  | (823) |  |  | (824) |  |  | (825) |  |  | (826) |  |  | (827) |  |  | (828) |  |  | (829) |  |  | (830) |  |  | (831) |  |  | (832) |  |  | (833) |  |  | (834) |  |  | (835) |  |  | (836) |  |  | (837) |  |  | (838) |  |  | (839) |  |  | (840) |  |  | (841) |  |  | (842) |  |  | (843) |  |  | (844) |  |  | (845) |  |  | (846) |  |  | (847) |  |  | (848) |  |  | (849) |  |  | (850) |  |  | (851) |  |  | (852) |  |  | (853) |  |  | (854) |  |  | (855) |  |  | (856) |  |  | (857) |  |  | (858) |  |  | (859) |  |  | (860) |  |  | (861) |  |  | (862) |  |  | (863) |  |  | (864) |  |  | (865) |  |  | (866) |  |  | (867) |  |  | (868) |  |  | (869) |  |  | (870) |  |  | (871) |  |  | (872) |  |  | (873) |  |  | (874) |  |  | (875) |  |  | (876) |  |  | (877) |  |  | (878) |  |  | (879) |  |  | (880) |  |  | (881) |  |  | (882) |  |  | (883) |  |  | (884) |  |  | (885) |  |  | (886) |  |  | (887) |  |  | (888) |  |  | (889) |  |  | (890) |  |  | (891) |  |  | (892) |  |  | (893) |  |  | (894) |  |  | (895) |  |  | (896) |  |  | (897) |  |  | (898) |  |  | (899) |  |  | (900) |  |  | (901) |  |  | (902) |  |  | (903) |  |  | (904) |  |  | (905) |  |  | (906) |  |  | (907) |  |  | (908) |  |  | (909) |  |  | (910) |  |  | (911) |  |  | (912) |  |  | (913) |  |  | (914) |  |  | (915) |  |  | (916) |  |  | (917) |  |  | (918) |  |  | (919) |  |  | (920) |  |  | (921) |  |  | (922) |  |  | (923) |  |  | (924) |  |  | (925) |  |  | (926) |  |  | (927) |  |  | (928) |  |  | (929) |  |  | (930) |  |  | (931) |  |  | (932) |  |  | (933) |  |  | (934) |  |  | (935) |  |  | (936) |  |  | (937) |  |  | (938) |  |  | (939) |  |  | (940) |  |  | (941) |  |  | (942) |  |  | (943) |  |  | (944) |  |  | (945) |  |  | (946) |  |  | (947) |  |  | (948) |  |  | (949) |  |  | (950) |  |  | (951) |  |  | (952) |  |  | (953) |  |  | (954) |  |  | (955) |  |  | (956) |  |  | (957) |  |  | (958) |  |  | (959) |  |  | (960) |  |  | (961) |  |  | (962) |  |  | (963) |  |  | (964) |  |  | (965) |  |  | (966) |  |  | (967) |  |  | (968) |  |  | (969) |  |  | (970) |  |  | (971) |  |  | (972) |  |  | (973) |  |  | (974) |  |  | (975) |  |  | (976) |  |  | (977) |  |  | (978) |  |  | (979) |  |  | (980) |  |  | (981) |  |  | (982) |  |  | (983) |  |  | (984) |  |  | (985) |  |  | (986) |  |  | (987) |  |  | (988) |  |  | (989) |  |  | (990) |  |  | (991) |  |  | (992) |  |  | (993) |  |  | (994) |  |  | (995) |  |  | (996) |  |  | (997) |  |  | (998) |  |  | (999) |  |  | (1000) |  |  | (1001) |  |  | (1002) |  |  | (1003) |  |  | (1004) |  |  | (1005) |  |  | (1006) |  |  | (1007) |  |  | (1008) |  |  | (1009) |  |  | (1010) |  |  | (1011) |  |  | (1012) |  |  | (1013) |  |  | (1014) |  |  | (1015) |  |  | (1016) |  |  | (1017) |  |  | (1018) |  |  | (1019) |  |  | (1020) |  |  | (1021) |  |  | (1022) |  |  | (1023) |  |  | (1024) |  |  | (1025) |  |  | (1026) |  |  | (1027) |  |  | (1028) |  |  | (1029) |  |  | (1030) |  |  | (1031) |  |  | (1032) |  |  | (1033) |  |  | (1034) |  |  | (1035) |  |  | (1036) |  |  | (1037) |  |  | (1038) |  |  | (1039) |  |  | (1040) |  |  | (1041) |  |  | (1042) |  |  | (1043) |  |  | (1044) |  |  | (1045) |  |  | (1046) |  |  | (1047) |  |  | (1048) |  |  | (1049) |  |  | (1050) |  |  | (1051) |  |  | (1052) |  |  | (1053) |  |  | (1054) |  |  | (1055) |  |  | (1056) |  |  | (1057) |  |  | (1058) |  |  | (1059) |  |  | (1060) |  |  | (1061) |  |  | (1062) |  |  | (1063) |  |  | (1064) |  |  | (1065) |  |  | (1066) |  |  | (1067) |  |  | (1068) |  |  | (1069) |  |  | (1070) |  |  | (1071) |  |  | (1072) |  |  | (1073) |  |  | (1074) |  |  | (1075) |  |  | (1076) |  |  | (1077) |  |  | (1078) |  |  | (1079) |  |  | (1080) |  |  | (1081) |  |  | (1082) |  |  | (1083) |  |  | (1084) |  |  | (1085) |  |  | (1086) |  |  | (1087) |  |  | (1088) |  |  | (1089) |  |  | (1090) |  |  | (1091) |  |  | (1092) |  |  | (1093) |  |  | (1094) |  |  | (1095) |  |  | (1096) |  |  | (1097) |  |  | (1098) |  |  | (1099) |  |  | (1100) |  |  | (1101) |  |  | (1102) |  |  | (1103) |  |  | (1104) |  |  | (1105) |  |  | (1106) |  |  | (1107) |  |  | (1108) |  |  | (1109) |  |  | (1110) |  |  | (1111) |  |  | (1112) |  |  | (1113) |  |  | (1114) |  |  | (1115) |  |  | (1116) |  |  | (1117) |  |  | (1118) |  |  | (1119) |  |  | ( |  |

## K Declaration of AI Use

In accordance with the University’s guidance on generative AI (“Use of GenAI is permitted to support assignments”), this appendix declares the use of AI tools throughout the dissertation. All use was conducted with human oversight and control; AI-generated outputs were reviewed, modified, and independently verified before inclusion. No AI-generated text was submitted without such verification. This declaration follows the order of permitted activities specified in the assessment guidelines. AI-assisted activities not explicitly listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) fall under the same oversight and verification standards described in the relevant rows above.

**Table 26.** Declaration of AI use by permitted activity.

| Activity                                    | AI tool(s)                                                                               | How output was used                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Researching and refining ideas              | ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)                              | ChatGPT was used iteratively to develop ideas that aligned with the author’s intent. Coding agents independently checked data and code availability to confirm feasibility. All final decisions and conclusions are the author’s own. Final research design was formally presented to and approved by supervisor.              |
| Information retrieval / background research | Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)                               | AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.                                                                                                                                                       |
| Drafting an outline / organising thoughts   | MiniMax / DeepSeek (via OpenCode), GPT (via Codex)                                       | Generated code scaffolding and assisted with the full analysis pipeline (embedding, scoring, validation, robustness, appendix outputs). The centroid-assignment step was verified against hand-labeled samples, and the scoring and coverage-gap functions were unit-tested. Analytical decisions were directed by the author. |
| Refining research questions                 | ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)                              | Overlaps with row 1. ChatGPT sharpened framing based on reviewed literature; coding agents confirmed questions were answerable with available data and models. The author retained full control over the final formulation. RQs presented to and approved by supervisor.                                                       |
| Checking spelling and grammar               | ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex) | Coding agents provided inline stylistic and phrasing suggestions during drafting. Web and mobile versions of ChatGPT, Gemini, and Claude were used for editorial review and proofreading. All suggestions were accepted or rejected per standard editorial judgment.                                                           |